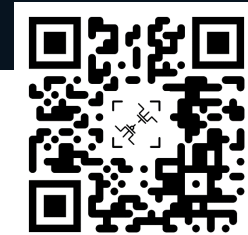




# Design and Simulation of RF Integrated Circuits with Open-Source CAD Tools and Process Design Kits

**SLIDES >>>>>>>>>**



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# Tutorial Outline

- **Part 1:**

**Advanced Compact MOSFET Model 2 - ACM2 model**

- **Part 2:**

**Implementing the ACM2 model**

- **Part 3:**

**ACM2 - Parameter extraction**

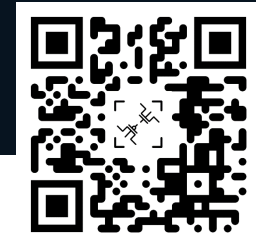
- **Part 4:**

**Design Methodology-Application to Low Noise Amplifier**



# Part 1

## Advanced Compact MOSFET Model: ACM2



"Scan me"



Carlos Galup-Montoro

[https://github.com/ACMmodel/MOSFET\\_model](https://github.com/ACMmodel/MOSFET_model)

# Outline

- **Introduction: ACM timeline**
- **ACM2 model**
- **Long-channel:  $I_D$  and  $g_m/I_D$  models**



# ACM timeline

- 1993  $\phi s$ -based model (SBMICRO, Campinas, Br)
- 1995 Long-channel charge-based model (SSE, Nov.)
- 1998 Most referenced ACM paper (JSSC, Oct.)
- 2000 ACM model in SMASH simulator ( CICC, Orlando)

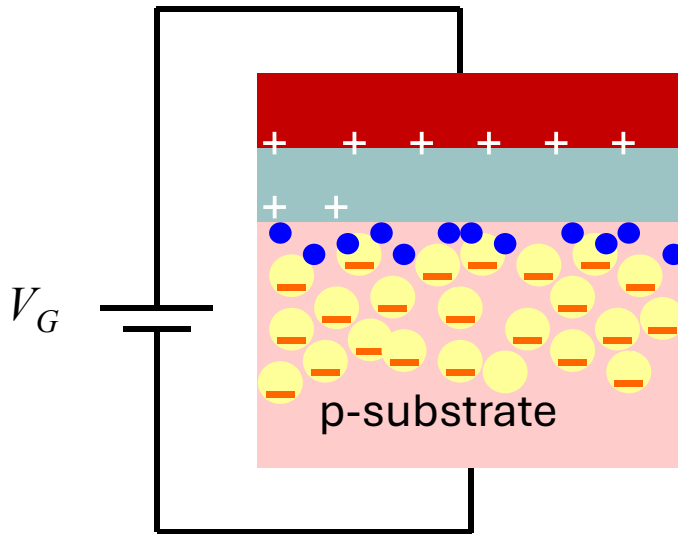


- 2018 Short course on ACM for ST organized by Prof. Silvain at Grenoble
- 2021 4-parameter single-piece model (NorCAS, Oslo)
- 2023 ACM2 in VERILOG-AMS ( NEWCAS, Edinburgh)

# Outline

- Introduction: ACM timeline
- **ACM2 model**
- Long-channel:  $I_D$  and  $g_m/I_D$  models

# The capacitive model of the MOSFET



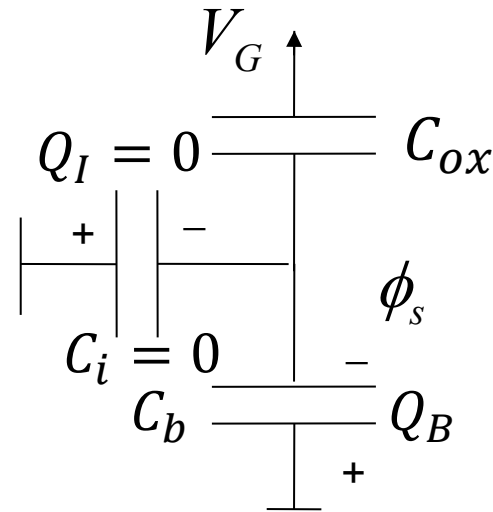
$\phi_s$  surface potential

$C_{ox}$  oxide capacitance per unit area

$C_b$  depletion capacitance per unit area

$Q_I$  carrier charge density

$V_{T0}$  threshold voltage



$$\frac{\Delta\phi_s}{\Delta V_G} = \frac{C_{ox}}{C_{ox} + C_b} = \frac{1}{n}$$

$$\phi_s = 2\phi_F + \frac{V_G - V_{T0}}{n} = 2\phi_F + V_P$$

# ACM2 current law

From 3 approximations: normalized current vs. normalized charge densities at source and drain



$$i_D = \frac{(q_S + q_D + 2)}{1 + \zeta(q_S - q_D)} (q_S - q_D)$$

$$i_D = I_D/I_S \quad I_S = \frac{W}{L} \mu_s n C_{ox} \frac{\phi_t^2}{2}$$

normalization (specific) current

$$q_{S(D)} = Q_{S(D)} / (-n C_{ox} \phi_t)$$

$-n C_{ox} \phi_t$  thermal charge

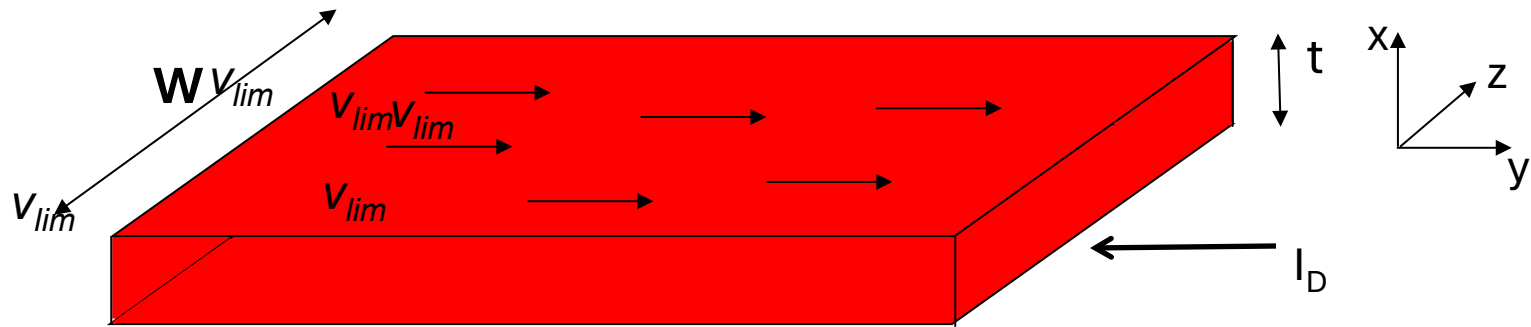
$$SI : q_{S(D)} \gg 1 \quad WI : q_{S(D)} \ll 1$$

Short-channel parameter  $\zeta$  :

$$\zeta = \frac{(\mu_s \phi_t / L)}{v_{lim}}$$

ratio of diffusion-related velocity to saturation velocity

# Physics-based saturation



Saturation current due to saturation velocity of the carriers

$$I_{Dsat} = -W Q_{Dsat} v_{lim}$$

$Q_{Dsat}$  is the saturation inversion charge per unit area

or, using normalized variables

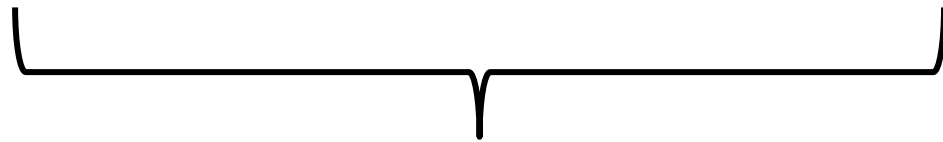
$$i_{Dsat} = \frac{2}{\zeta} q_{dsat}$$

“Carrier velocity approaches  $v_{sat}$ , but never reaches  $v_{sat}$ ”

Y.Taur TED March 2019

# Physics-based saturation: design model

$$i_{Dsat} = \frac{2}{\zeta} q_{dsat} \qquad i_{Dsat} = \frac{(q_s + q_{Dsat} + 2)}{1 + \zeta(q_s - q_{Dsat})} (q_s - q_{Dsat})$$



$$q_{Dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

**or equivalently**

$$q_s = \sqrt{1 + \frac{2}{\zeta} q_{dsat}} - 1 + q_{dsat}$$

# The Unified Charge Control Model (UCCM)

$$\frac{V_P - V_C}{\phi_t} = q_I - 1 + \ln q_I$$

$$V_S \leq V_C \leq V_D$$

$$q_S \leq q_I \leq q_D$$



$$\frac{V_{DS}}{\phi_t} = q_S - q_D + \ln \frac{q_S}{q_D}$$

The “Regional” Weak (WI) and Strong Inversion (SI) Approximations

**WI**

**SI**

$$q_I \ll 1 \rightarrow V_P - V_C \ll -\phi_t$$

$$q_I \gg 1 \rightarrow V_P - V_C \gg \phi_t$$

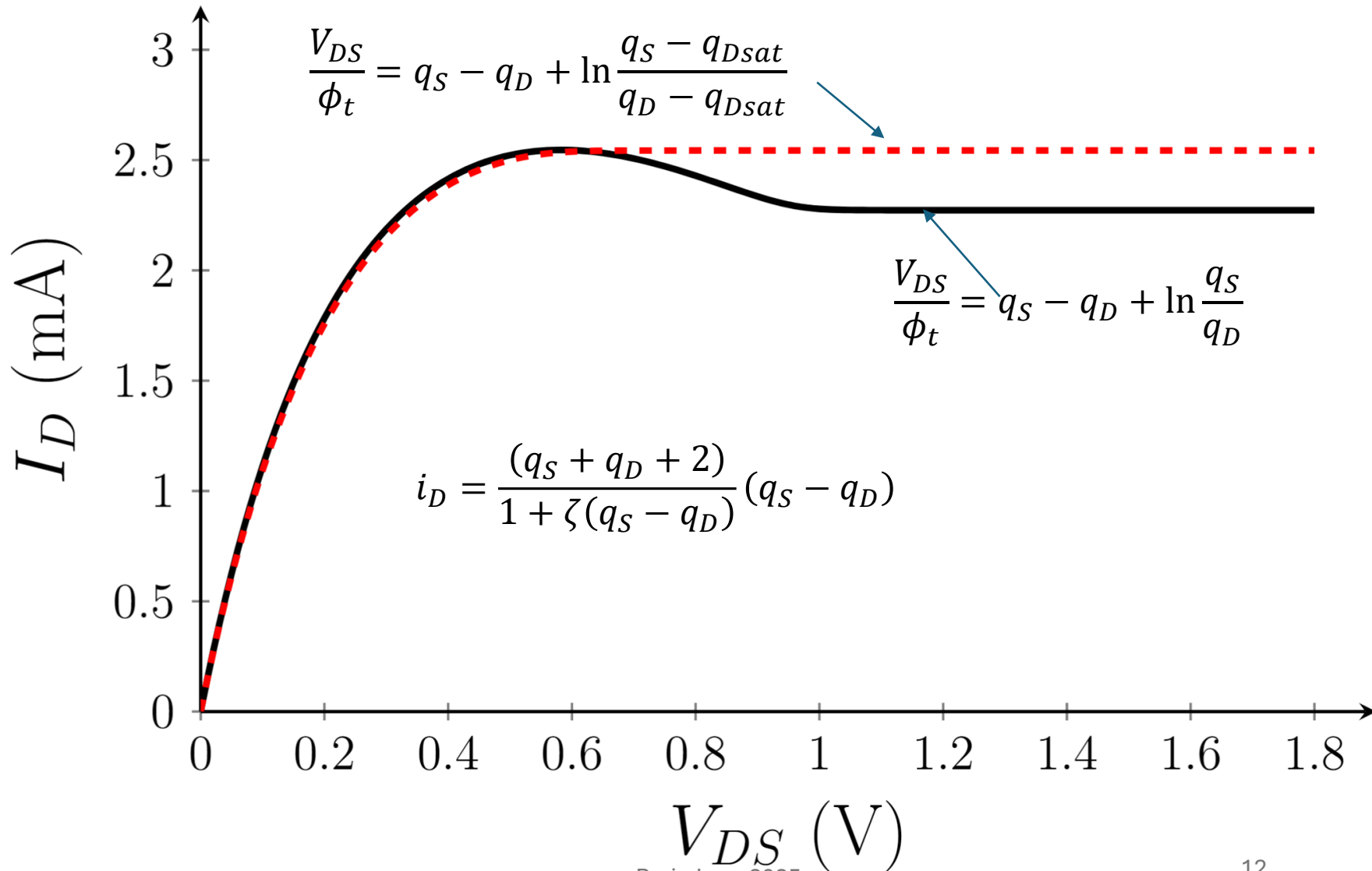
$$q_I \cong e^{\frac{V_P - V_C + \phi_t}{\phi_t}}$$

$$q_I \cong \frac{V_P - V_C + \phi_t}{\phi_t}$$

Error <10% for  $q_I < 0.22$

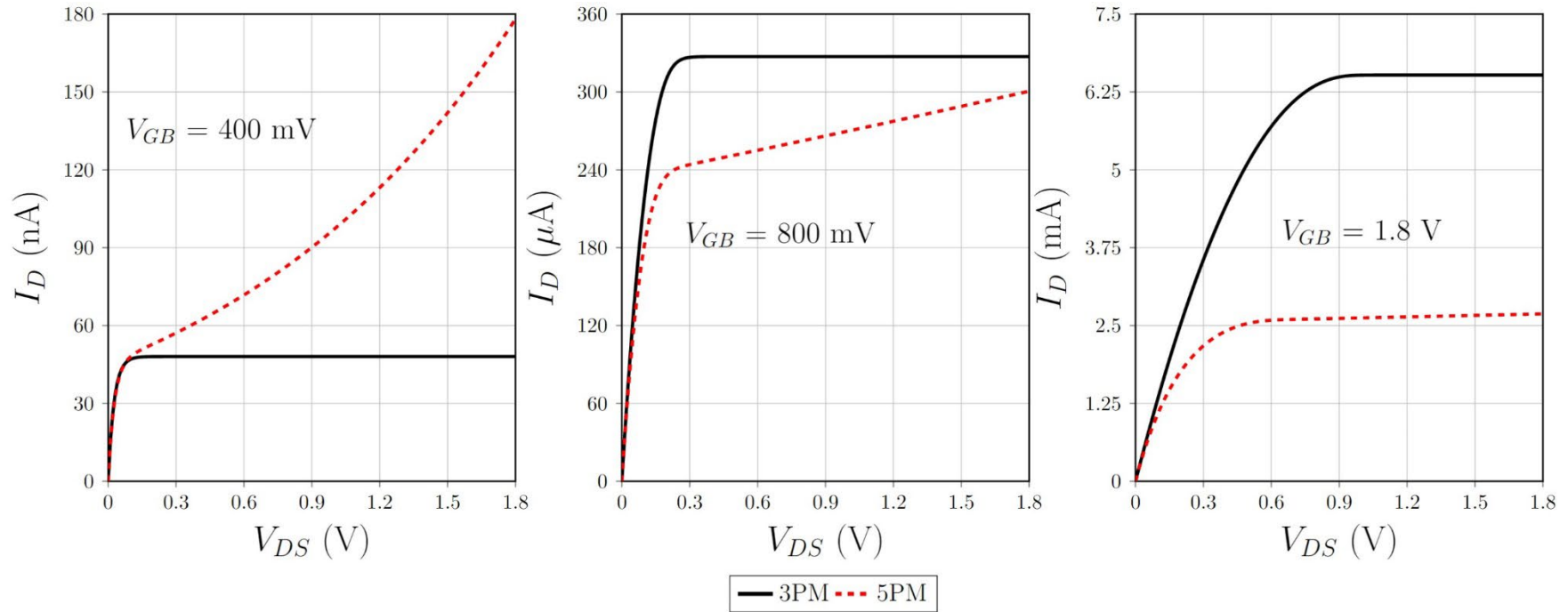
Error <10% for  $q_I > 20$

# UCCM including the effect of velocity saturation





# Output curves including DIBL and $v_{sat}$



*DIBL model:*  $V_T = V_{T0} - \sigma(V_S + V_D)$

| Transistor | $W/L$<br>( $\mu\text{m}/\mu\text{m}$ ) | $V_{T0}$ (mV) | $I_S$ ( $\mu\text{A}$ ) | n    | $\sigma$ | $\zeta$ |
|------------|----------------------------------------|---------------|-------------------------|------|----------|---------|
| NMOS2V     | 5/0.18                                 | 528           | 5.52                    | 1.37 | 0.025    | 0.056   |

# Outline

- Introduction: ACM timeline
- ACM2 model
- Long-channel:  $I_D$  and  $g_m/I_D$  models

# Long-channel charge-based model

For short channel MOS

$$I_D = I_S \frac{(q_S + q_D + 2)}{1 + \zeta(q_S - q_D)} (q_S - q_D)$$

$$\frac{V_P - V_S}{\phi_t} = q_S - 1 + \ln q_S \quad \frac{V_{DS}}{\phi_t} = q_S - q_D + \ln \frac{q_S - q_{Dsat}}{q_D - q_{Dsat}}$$

For long channel MOS  $\zeta=0 \rightarrow q_{Dsat}=0$

$$I_D = I_S(q_S + q_D + 2)(q_S - q_D) = I_S[(q_S^2 + 2q_S) - (q_D^2 + 2q_D)]$$

$$\frac{V_P - V_{S(D)}}{\phi_t} = q_{S(D)} - 1 + \ln q_{S(D)}$$

Obs:  $\frac{dq_{S(D)}}{dV_G} = -\frac{1}{n} \frac{dq_{S(D)}}{dV_{S(D)}} \quad \frac{dq_D}{dV_D} = -\frac{1}{\phi_t} \frac{q_D}{q_D + 1} \quad \frac{dI_D}{dq_D} = -2I_S(q_D + 1)$

# Small-signal transconductances

$$g_{md} = \frac{dI_D}{dV_D} = \frac{dI_D}{dq_D} \frac{dq_D}{dV_D} = 2I_S(q_D + 1) \frac{1}{\phi_t} \frac{q_D}{q_D + 1} = \frac{2I_S}{\phi_t} q_D$$

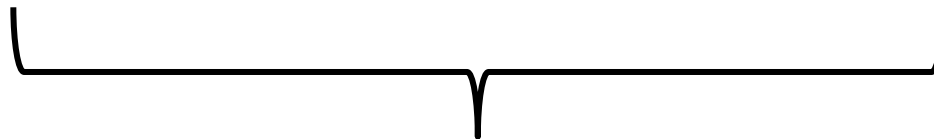
Symmetry



$$g_{ms} = \frac{2I_S}{\phi_t} q_S$$

**Amazingly simple  
(see Appendix)**

$$g_m = \frac{dI_D}{dV_G} = \frac{dI_D}{dq_S} \frac{dq_S}{dV_G} + \frac{dI_D}{dq_D} \frac{dq_D}{dV_G} \qquad \frac{dq_{S(D)}}{dV_G} = -\frac{1}{n} \frac{dq_{S(D)}}{dV_{S(D)}}$$



$$g_m = \frac{g_{ms} - g_{md}}{n}$$

$$g_m = \frac{g_{ms}}{n} \longrightarrow \text{in saturation}$$

# Unified Current Control Model (UICM)-I

$$I_D = I_S[(q_S^2 + 2q_S) - (q_D^2 + 2q_D)] \quad (A)$$

(A) can also be written as

$$I_D = I_F - I_R = I_S[i_f - i_r] \quad (B)$$

$I_F, I_R$  : forward and reverse currents

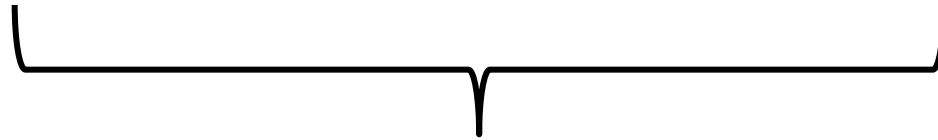
$i_{f(r)} = q_{S(D)}^2 + 2q_{S(D)}$  : forward (reverse) inversion coefficients

$$\downarrow$$
$$q_{S(D)} = \sqrt{1 + i_{f(r)}} - 1$$

# Unified Current Control Model (UICM)-II

$$\frac{V_P - V_{S(D)}}{\phi_t} = q_{S(D)} - 1 + \ln q_{S(D)}$$

$$q_{S(D)} = \sqrt{1 + i_{f(r)}} - 1$$



## Normalized UICM

$$\frac{V_P - V_{S(D)}}{\phi_t} = \sqrt{1 + i_{f(r)}} - 2 + \ln \left( \sqrt{1 + i_{f(r)}} - 1 \right)$$



$$\frac{V_{DS}}{\phi_t} = q_S - q_D + \ln \frac{q_S}{q_D} = \sqrt{1 + i_f} - \sqrt{1 + i_r} + \ln \left( \frac{\sqrt{1 + i_f} - 1}{\sqrt{1 + i_r} - 1} \right)$$

$$g_m/I_D$$

$$g_{ms(d)} = \frac{2I_S}{\phi_t} \left( \sqrt{1 + i_{f(r)}} - 1 \right) = \frac{W}{L} \mu C_{ox} n \phi_t \left( \sqrt{1 + i_{f(r)}} - 1 \right)$$

$$g_m = \frac{g_{ms} - g_{md}}{n}$$

$$\frac{g_m}{I_D} = \frac{2}{n \phi_t (\sqrt{1 + i_f} + \sqrt{1 + i_r})}$$

For  $V_{DS}/\phi_t \ll 1$  we have  $i_f \approx i_r$

In saturation  $i_f \gg i_r$

$$\frac{g_m}{I_D} \cong \frac{1}{n \phi_t \sqrt{1 + i_f}}$$

$$\frac{g_m}{I_D} \cong \frac{2}{n \phi_t (\sqrt{1 + i_f} + 1)}$$

# REFERENCES

- A. I. A. Cunha, M. C. Schneider and C. Galup-Montoro, "Derivation of the Unified Charge Control Model and Parameter Extraction Procedure", Solid-State Electronics, March 1999.
- O. C. Gouveia Filho, A. I. A. Cunha, M. C. Schneider and C. Galup-Montoro, "Advanced compact model for short-channel MOS transistors", IEEE Custom Integrated Circuits Conference, Orlando, FL, USA, May 2000.
- C. Galup-Montoro and M. C. Schneider, *MOSFET Modeling for Circuit Analysis and Design*, World Scientific, 2007.
- C. M. Adornes, D. G. Alves Neto, M. C. Schneider and C. Galup-Montoro, "Bridging the Gap between Design and Simulation of Low-Voltage CMOS Circuits", Journal of Low Power Electronics and Applications, June 2022.
- D. G. Alves Neto *et al*, "Design-oriented single-piece 5-DC parameter MOSFET model," *IEEE Access*, 2024.



# Appendix: the exact model of the long-channel MOSFET<sup>1</sup>

$$I_D = -\frac{W}{L} \int_{V_S}^{V_D} \mu Q_I(V_C) dV_C$$

Consequently, the **exact** expressions for  $g_{ms}$  and  $g_{md}$  are

$$g_{md} = \frac{dI_D}{dV_D} = -\frac{W}{L} \mu Q_D = \frac{2I_S}{\phi_t} q_D$$

$$g_{ms} = -\frac{dI_D}{dV_S} = -\frac{W}{L} \mu Q_S = \frac{2I_S}{\phi_t} q_S$$

<sup>1</sup> H. C. Pao and C. T. Sah, 'Effects of diffusion current on characteristics of metal-oxide (insulator)-semiconductor transistors' Solid-State Electronics, Oct 1966



## Part 2:

# Implementing the ACM2 model



Manuel J. Barragán

[https://github.com/ACMmodel/MOSFET\\_model](https://github.com/ACMmodel/MOSFET_model)



"Scan me"



- ❑ The ACM2 model (a quick recap)
- ❑ Numerical solving of the UCCM equations
- ❑ acm 433 algorithm and Lambert function
- ❑ Analytical approximation of Lambert functions
- ❑ Towards a closed-form I-V MOSFET model

# The ACM2 model (a quick recap)

## The 5-parameter ACM2 model

- Only 5 parameters:  $n$ ,  $I_{S0}$ ,  $V_{T0}$ ,  $\sigma$  and  $\zeta$

$$I_D = I_{S0} \frac{q_S + q_D + 2}{1 + \zeta |q_S - q_D|} (q_S - q_D)$$

where

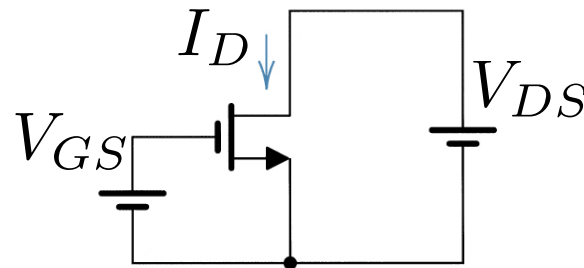
$$\frac{V_P - V_{SB}}{U_T} = q_S - 1 + \ln q_S$$

$$\frac{V_{DS}}{U_T} = q_S - q_D + \ln \frac{q_S - q_{Dsat}}{q_D - q_{Dsat}}$$

with

$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

$$q_{Dsat} = q_S + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_S}{\zeta}}$$



# The ACM2 model (a quick recap)

## The 5-parameter ACM2 model

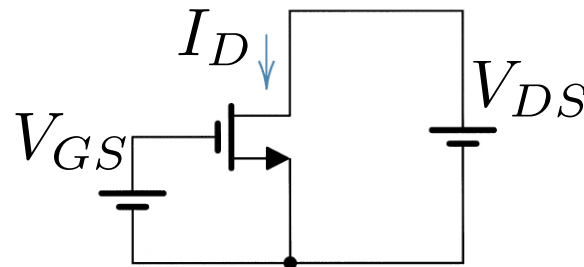
- Only 5 parameters:  $n$ ,  $I_{S0}$ ,  $V_{T0}$ ,  $\sigma$  and  $\zeta$

$$I_D = I_{S0} \frac{q_S + q_D + 2}{1 - \zeta |q_S - q_D|} (q_S - q_D)$$

where

$$\frac{V_P - V_{SB}}{U_T} = q_S - 1 + \ln q_S$$

$$\frac{V_{DS}}{U_T} = q_S - q_D + \ln \frac{q_S - q_{Dsat}}{q_D - q_{Dsat}}$$



with

$$V_P = \frac{V_{GB} - V_{T0} + \frac{\sigma(V_{DB} + V_{SB})}{n}}{n}$$

$$q_{Dsat} = q_S + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_S}{\zeta}}$$

# The ACM2 model (a quick recap)

## Computing $I_D$ in practice:

Given the model parameters and polarization voltages

$n, I_{S0}, V_{T0}, \sigma$  and  $\zeta$

$V_G, V_D, V_S, V_B$

Step 1: Compute  $V_P$ : 
$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

Step 2: Compute  $q_S$ : 
$$\frac{V_P - V_{SB}}{U_T} = q_S - 1 + \ln q_S$$

Step 3: Compute  $q_{Dsat}$ : 
$$q_{Dsat} = q_S + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_S}{\zeta}}$$

Step 4: Compute  $q_D$ : 
$$\frac{V_{DS}}{U_T} = q_S - q_D + \ln \frac{q_S - q_{Dsat}}{q_D - q_{Dsat}}$$

Step 5: Compute  $I_D$ : 
$$I_D = I_{S0} \frac{q_S + q_D + 2}{1 + \zeta |q_S - q_D|} (q_S - q_D)$$



# The ACM2 model (a quick recap)

## Computing $I_D$ in practice:

Given the model parameters and polarization voltages

$n, I_{S0}, V_{T0}, \sigma$  and  $\zeta$

$V_G, V_D, V_S, V_B$

Step 1: Compute  $V_P$ : 
$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

Step 2: Compute  $q_S$ : 
$$\frac{V_P - V_{SB}}{U_T} = q_S - 1 + \ln q_S$$

Step 3: Compute  $q_{Dsat}$ : 
$$q_{Dsat} = q_S + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_S}{\zeta}}$$

Step 4: Compute  $q_D$ : 
$$\frac{V_{DS}}{U_T} = q_S - q_D + \ln \frac{q_S - q_{Dsat}}{q_D - q_{Dsat}}$$

Step 5: Compute  $I_D$ : 
$$I_D = I_{S0} \frac{q_S + q_D + 2}{1 + \zeta |q_S - q_D|} (q_S - q_D)$$



Transcendental  
equations

# acm 443 Algorithm and Lambert W Functions

How to solve  $\frac{V_P - V_{SB}}{U_T} = q_S - 1 + \ln q_S$  efficiently?

$$\frac{V_P - V_{SB}}{U_T} = q_S - 1 + \ln q_S \quad \Rightarrow \quad e^{\left(\frac{V_P - V_{SB}}{U_T} - 1\right)} = q_S e^{q_S} \quad \Rightarrow \quad x = w e^w$$

Algorithm 443: Solution of the transcendental equation  $w e^w = x$

$$w_n = \begin{cases} \frac{x + \frac{4}{3}x^2}{1 + \frac{7}{3}x + \frac{5}{6}x^2}, & x < 0.7385 \\ \ln(x) - \frac{24(\ln(x)\ln(x) + 2\ln(x) - 3)}{7\ln(x)\ln(x) + 58\ln(x) + 127}, & x \geq 0.7385 \end{cases}$$
$$z_n = \ln(x) - w_n - \ln(w_n)$$
$$e_n = \frac{z_n}{1 + w_n} \frac{2(1 + w_n) \left(1 + w_n + \frac{2}{3}z_n\right) - z_n}{2(1 + w_n) \left(1 + w_n + \frac{2}{3}z_n\right) - 2z_n}$$
$$w = w_n(1 + e_n)$$



# acm 443 Algorithm and Lambert W Functions

How to solve  $\frac{V_{DS}}{U_T} = q_S - q_D + \ln \frac{q_S - q_{Dsat}}{q_D - q_{Dsat}}$  efficiently?

$$\frac{V_{DS}}{U_T} = q_S - q_{Dsat} - (q_D - q_{Dsat}) + \ln \frac{q_S - q_{Dsat}}{q_D - q_{Dsat}} \quad \rightarrow$$

$$\rightarrow (q_S - q_{Dsat}) \frac{e^{(q_S - q_{Dsat})}}{e^{V_{DS}/U_T}} = (q_D - q_{Dsat}) e^{(q_D - q_{Dsat})} \quad \rightarrow x = w e^w$$

Algorithm 443

$$w_n = \begin{cases} \frac{x + \frac{4}{3}x^2}{1 + \frac{7}{3}x + \frac{5}{6}x^2}, & x < 0.7385 \\ \ln(x) - \frac{24(\ln(x)\ln(x) + 2\ln(x) - 3)}{7\ln(x)\ln(x) + 58\ln(x) + 127}, & x \geq 0.7385 \end{cases}$$
$$z_n = \ln(x) - w_n - \ln(w_n)$$
$$e_n = \frac{z_n}{1 + w_n} \frac{2(1 + w_n) \left(1 + w_n + \frac{2}{3}z_n\right) - z_n}{2(1 + w_n) \left(1 + w_n + \frac{2}{3}z_n\right) - 2z_n}$$
$$w = w_n(1 + e_n)$$
$$q_D = w + q_{Dsat}$$

# acm 443 Algorithm and Lambert W Functions

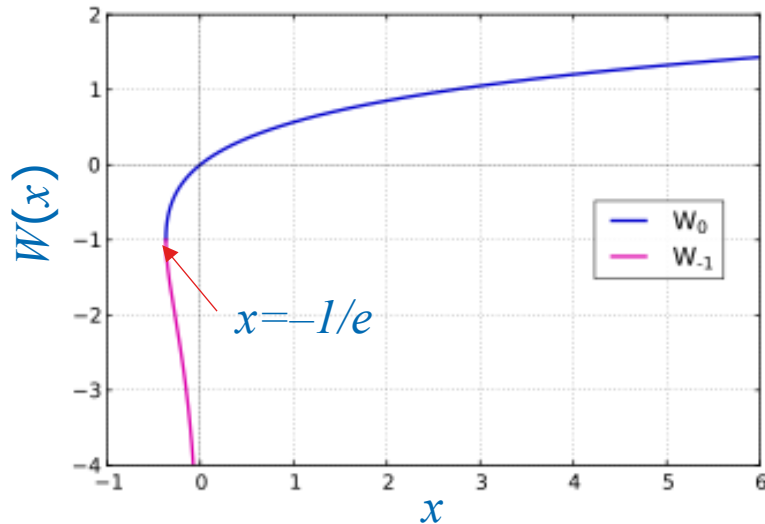
Can we consider the problem analytically?

- The Lambert  $W$  function  $z = w e^w$  for  $w$  any complex number was studied by Johann Lambert in 1758, then properly defined by L. Euler in 1783

# acm 443 Algorithm and Lambert W Functions

Can we consider the problem analytically?

- The Lambert  $W$  function  $z = w e^w$  for  $w$  any complex number was studied by Johann Lambert in 1758, then properly defined by L. Euler in 1783
- For real numbers  $x$  and  $W(x)$ , equation  $x = W(x) e^{W(x)}$  has solutions  $W_0$  and  $W_{-1}$ :



$q_S$ ,  $q_D$  and  $q_{Dsat}$ , and  $I_D$  can be expressed as a function of  $V_G$ ,  $V_D$ ,  $V_S$  using  $W_0$

## Expressing $I_D$ using Lamber W function

$$I_D = I_{S0} \frac{q_S + q_D + 2}{1 + \zeta |q_S - q_D|} (q_S - q_D)$$

$$q_S = W_0 \left( e^{\left( \frac{V_P - V_{SB}}{U_T} - 1 \right)} \right)$$

$$q_D = q_{Dsat} + W_0 \left( \left( W_0 \left( e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) - q_{Dsat} \right) e^{\left( W_0 \left( e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) - q_{Dsat} \right) - \frac{V_{DS}}{U_T}} \right)$$

$$q_{Dsat} = W_0 \left( e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) + 1 + \frac{1}{\zeta} - \sqrt{\left( 1 + \frac{1}{\zeta} \right)^2 + \frac{2W_0 \left( e^{\frac{V_P - V_{SB}}{U_T} - 1} \right)}{\zeta}}$$

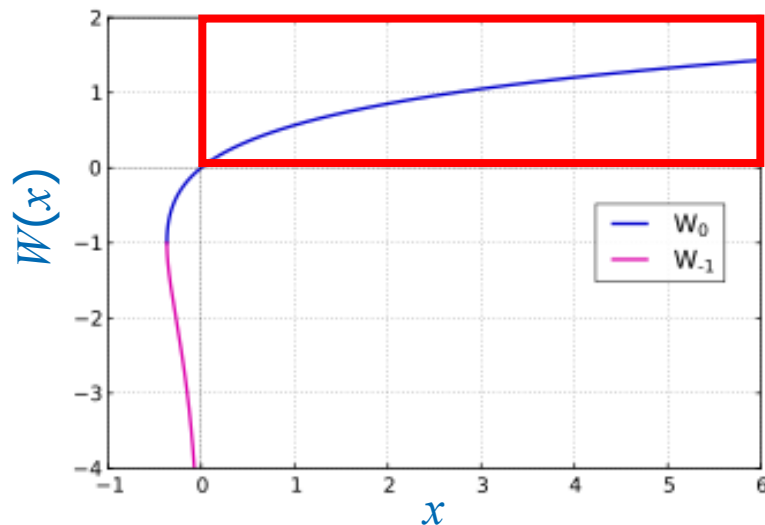
$W_0$  is available in main programming languages: Matlab (lambertw), Octave (lambertw), Python (lambertw in scipy lib), R (lambertW0), ...

...but can we approximate  $W_0$  in terms of elementary functions?

# acm 443 Algorithm and Lambert W Functions

## Approximating the Lambert $W$ function

For real numbers  $x$  and  $W(x)$ ,  $x = W(x) e^{W(x)}$



Approximation of  $W_0(x)$  for  $x > 0$

$$W_{0,0}(x) = \ln [1 + \alpha(x)x]$$

$$W_{0,n}(x) = \frac{W_{n-1}(x)}{1 + W_{n-1}(x)} \left[ 1 + \ln \left( \frac{x}{W_{n-1}(x)} \right) \right]$$

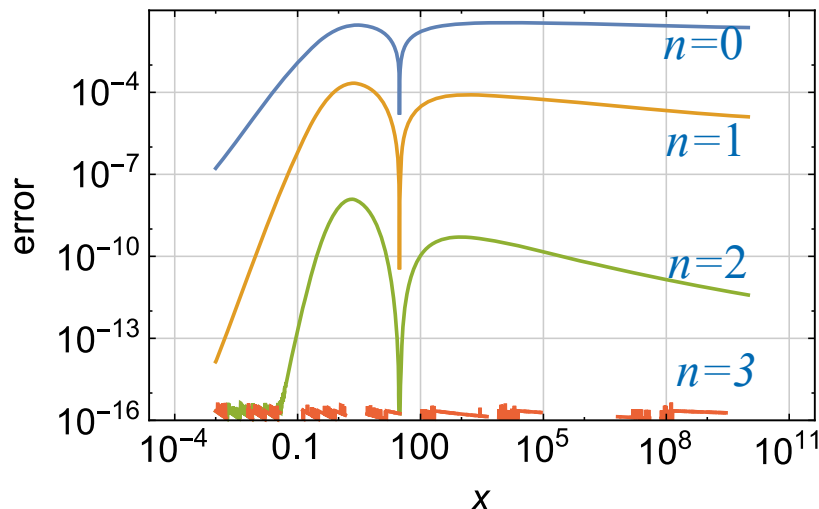
For  $n=1,2,3,\dots$

$$\text{where } \alpha(x) = \frac{1}{1 + \frac{\ln(1+x)}{2}}$$

# acm 443 Algorithm and Lambert W Functions

## Approximating the Lambert $W$ function

For real numbers  $x$  and  $W(x)$ ,  $x = W(x) e^{W(x)}$



Approximation of  $W_0(x)$  for  $x > 0$

$$W_{0,0}(x) = \ln[1 + \alpha(x)x]$$

$$W_{0,n}(x) = \frac{W_{n-1}(x)}{1 + W_{n-1}(x)} \left[ 1 + \ln \left( \frac{x}{W_{n-1}(x)} \right) \right]$$

For  $n=1,2,3,\dots$

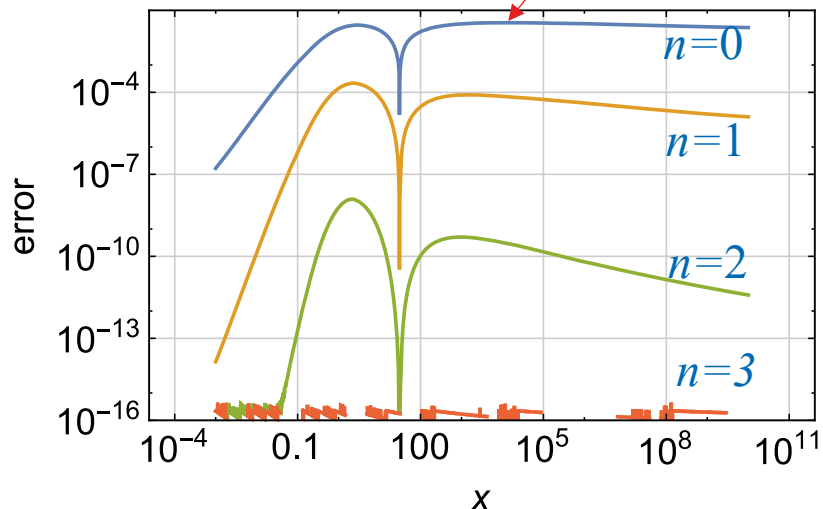
$$\text{where } \alpha(x) = \frac{1}{1 + \frac{\ln(1+x)}{2}}$$

# acm 443 Algorithm and Lambert W Functions

## Approximating the Lambert $W$ function

For real numbers  $x$  and  $W(x)$ ,  $x = W(x) e^{W(x)}$

Max. rel. error < 4%



Approximation of  $W_0(x)$  for  $x > 0$

$$W_{0,0}(x) = \ln [1 + \alpha(x)x]$$

where 
$$\alpha(x) = \frac{1}{1 + \frac{\ln(1+x)}{2}}$$

# acm 443 Algorithm and Lambert W Functions

## Approximating the Lambert $W$ function

### Approximating $q_S$

$$q_S = W_0 \left( e^{\left( \frac{V_P - V_{SB}}{U_T} - 1 \right)} \right) \quad \rightarrow \quad q_S \approx W_{0,0}(x) = \ln \left[ 1 + \frac{1}{1 + \frac{\ln[1 + e^{\left( \frac{V_P - V_{SB}}{U_T} - 1 \right)}]}{2}} e^{\left( \frac{V_P - V_{SB}}{U_T} - 1 \right)} \right]$$

### Approximating $q_D$ is similar from:

$$q_D = q_{Dsat} + W_0 \left( \left( W_0 \left( e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) - q_{Dsat} \right) e^{\left( W_0 \left( e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) - q_{Dsat} \right) - \frac{V_{DS}}{U_T}} \right)$$

$$q_{Dsat} = W_0 \left( e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) + 1 + \frac{1}{\zeta} - \sqrt{\left( 1 + \frac{1}{\zeta} \right)^2 + \frac{2W_0 \left( e^{\frac{V_P - V_{SB}}{U_T} - 1} \right)}{\zeta}}$$



# Towards an analytical I-V MOSFET model

Replacing  $q_S(V_G, V_S, V_D)$  and  $q_D(V_G, V_S, V_D)$  in  $I_D$  will give a closed-form I-V model... but the expression is complex

Proposal: modified empirical model targeting closed-form I-V

$$i_D = \frac{I_D}{I_{S0}} = \frac{(q_S - q_D)(q_S + q_D + 2)}{[1 + \zeta(q_S - q_D)][1 + \theta(q_S + q_D)]}$$

where

$$\frac{V_P - V_{S(D)B}}{U_T} = q_{S(D)} - q_{sat} - 1 + \ln(q_{S(D)} - q_{sat})$$

# Towards an analytical I-V MOSFET model

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where

$$q_{S(D)} = \ln \left[ 1 + \frac{e^{v_P - v_{S(D)B}} + 1}{1 + \frac{1}{2} \ln \left( 1 + e^{v_P - v_{S(D)B}} + 1 \right)} \right] \left[ 1 + \frac{\zeta}{2} \ln \left[ 1 + \frac{e^{v_P - v_{S(D)B}} + 1}{1 + \frac{1}{2} \ln \left( 1 + e^{v_P - v_{S(D)B}} + 1 \right)} \right] + \zeta \right]$$

# Take aways

- ❑ The ACM2 model contains transcendental equations
- ❑ Numerical solving and/or approximations are needed
- ❑ acm 433 algorithm and Lambert W function are two useful mathematical tools in this context
- ❑ Happy coding!!

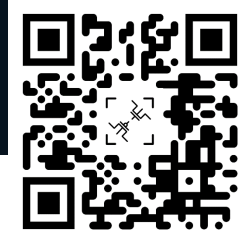


**Thank you!**



## Part 3

# Advanced Compact MOSFET Model: Parameter extraction



“Scan me”



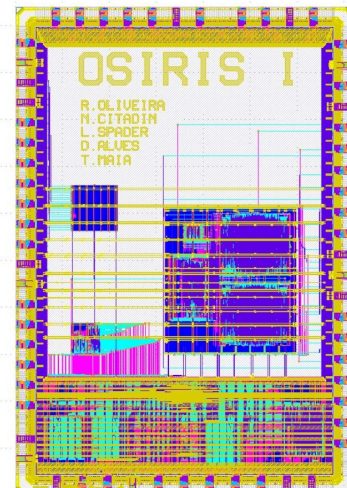
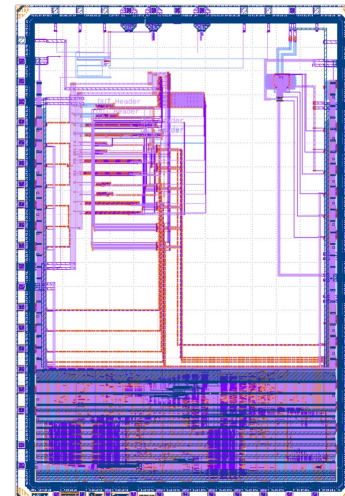
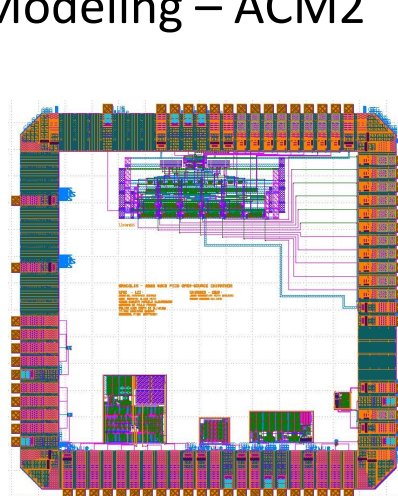
Deni Germano Alves Neto

[https://github.com/ACMmodel/MOSFET\\_model](https://github.com/ACMmodel/MOSFET_model)

# About me



- Universidade Federal de Santa Catarina – UFSC - Brazil
- Undergrad and Masters in IC design - 2022
  - Subject : Ultra-Low-Voltage IC circuits  $V_{DD} < 100\text{ mV}$ 
    - Dissertation: Ultra-Low-Voltage Standard Cell Library
      - ACM for low voltage circuits
- Contact with open-source IC design :
  - Chipathon - SSCS 2021 : Analog-front-end for Biosignals – AFEbio
  - Chipathon-SSCS & UNIC-CASS 2023/2024 – Analog/Digital IC design
- Start PhD in 2023 : MOSFET Modeling – ACM2
- Joint PhD between UFSC and
- UGA - Grenoble INP
  - TIMA (Currently based)
  - RFIC design



# Outline

- **ACM2 : Parameter Extraction**
  - 3-DC-parameter model
    - **$V_{T0}$ ,  $I_S$  and  $n$**
  - Transconductances in Saturation
    - Sigma & Zeta
  - DC and small-signals characteristics
  - Example: Inverter CMOS
- Overview of Open-source environment
  - Automatic parameter extraction

# ACM2: A simple 5-DC-parameter MOSFET model

$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

$$\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln(q_s)$$

Used to calculate  $q_s$

$$q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

$$\frac{V_{DS}}{\phi_t} = q_s - q_d + \ln\left(\frac{q_s - q_{dsat}}{q_d - q_{dsat}}\right)$$

Used to calculate  $q_d$

$$I_D = I_S \frac{(q_s + q_d + 2)}{1 + \zeta(q_s - q_d)} (q_s - q_d)$$

Specific  
current  
 $I_S$  ( $W, L$ )

Threshold  
voltage  
 $V_{T0}$  ( $W, L$ )

Slope  
factor  
 $n$  ( $W, L$ )

DIBL  
factor  
 $\sigma$  ( $W, L$ )

$V_{sat}$   
effect  
 $\zeta$  ( $W, L$ )



# $I_{s0}, V_{T0}$ and $n$ extraction $\rightarrow$ let's simplify and use only the 3-parameter model (no sat no DIBL)

## DC equations

$$I_D = I_S \frac{(q_s + q_d + 2)}{1 + \cancel{\zeta(q_s - q_d)}} (q_s - q_d) \quad \text{where} \quad I_S = \mu C_{ox} n \frac{\phi_t^2}{2} \frac{W}{L} = I_{SH} \frac{W}{L}$$

$$\frac{V_{DS}}{\phi_t} = q_s - q_D + \ln \left( \frac{q_s - \cancel{q_{Dsat}}}{q_D - \cancel{q_{Dsat}}} \right) \quad \text{No saturation effect} \rightarrow \text{no } \zeta, \text{no } q_{Dsat}$$

$$\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln(q_s)$$

$$V_P = \frac{V_{GB} - \cancel{V_{T0}} + \cancel{\sigma(V_{DB} + V_{SB})}}{n} \quad \text{No DIBL effect} \rightarrow \text{no } \sigma$$

Remember that:

$$I_D = I_S [i_f - i_r] \quad \text{with} \quad \begin{cases} q_s = \sqrt{1 + i_f} - 1 & \text{for forward current} \\ q_D = \sqrt{1 + i_r} - 1 & \text{for reverse current} \end{cases}$$

Then we re-write:

$$\frac{V_{DS}}{\phi_t} = \sqrt{1 + i_f} - \sqrt{1 + i_r} + \ln \left( \frac{\sqrt{1 + i_f} - 1}{\sqrt{1 + i_r} - 1} \right) \quad \frac{V_P - V_{SB}}{\phi_t} = \sqrt{1 + i_f} - 2 + \ln \left( \sqrt{1 + i_f} - 1 \right)$$

# $I_{S0}, V_{T0}$ and $n$ extraction $\rightarrow$ let's simplify and use only the 3-parameter model (no sat no DIBL)

## From DC equations

If we choose  $i_f = 3 \leftrightarrow q_S = 1 \leftrightarrow V_P = 0 \leftrightarrow V_{GB} = V_{T0}$

$$\frac{V_P - \cancel{V_{SB}}}{\phi_t} = \sqrt{1 + i_f} - 2 + \ln(\sqrt{1 + i_f} - 1)$$

If we add that we want to work at  $V_{DS} = \phi_t/2$

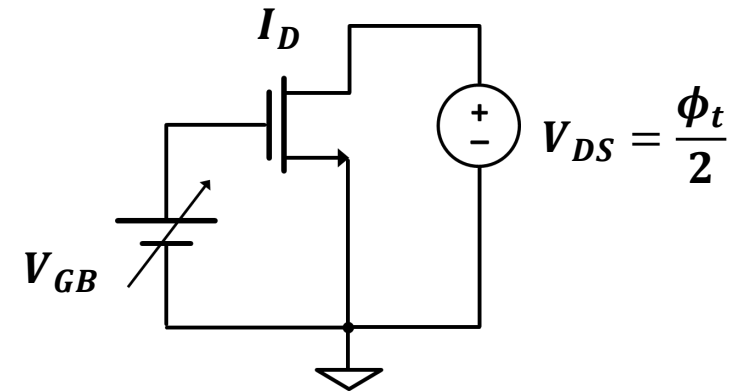
Since 
$$\frac{V_{DS}}{\phi_t} = \sqrt{1 + i_f} - \sqrt{1 + i_r} + \ln\left(\frac{\sqrt{1 + i_f} - 1}{\sqrt{1 + i_r} - 1}\right)$$

Then  $i_r = 2.12$  and  $I_D = I_{S0}[i_f - i_r] = 0.88 I_{S0}$

Consequence on small signal equations for  $i_f = 3$  and  $V_{DS} = \phi_t/2$

$$\frac{g_m}{I_D} = \frac{1}{n\phi_t} \cdot \frac{2}{(\sqrt{1 + i_f} + \sqrt{1 + i_r})} = \frac{0.531}{n\phi_t}$$

Meanwhile 
$$\left. \frac{g_m}{I_D} \right|_{max} = \frac{1}{n\phi_t} \quad i_f(i_r) \approx 0$$



# 3PM-ACM model in a nutshell

$$I_D = I_S [i_f - i_r] \quad \text{where} \quad I_S = \mu C_{ox} n \frac{\phi_t^2 W}{2 L} = I_{SH} \frac{W}{L}$$

$$\frac{V_P - V_{S(D)B}}{\phi_t} = \sqrt{1 + i_{f(r)}} - 2 + \ln \left( \sqrt{1 + i_{f(r)}} - 1 \right) \quad V_P \cong \frac{V_{GB} - V_{T0}}{n}$$

DC eqs

If we choose  $i_f = 3 \quad \Rightarrow \quad RHS = 0 \quad \Rightarrow \quad V_{GB} = V_{T0}$

$$\frac{V_{DS}}{\phi_t} = \sqrt{1 + i_f} - \sqrt{1 + i_r} + \ln \left( \frac{\sqrt{1 + i_f} - 1}{\sqrt{1 + i_r} - 1} \right)$$

$$g_{ms(d)} = \frac{2I_S}{\phi_t} \left( \sqrt{1 + i_{f(r)}} - 1 \right) \quad \Rightarrow \quad \frac{W}{L} = \frac{g_{ms(d)} \phi_t}{2I_{SH} (\sqrt{1 + i_{f(r)}} - 1)}$$

$$g_m = \frac{g_{ms} - g_{md}}{n}$$

$$\frac{g_m}{I_D} = \frac{d(\ln I_D)}{dV_G} = \frac{2}{n \phi_t (\sqrt{1 + i_f} + \sqrt{1 + i_r})}$$

Small-signal eqs

# $I_S, V_{T0}$ and $n$ extraction

## The $g_m/I_D$ method

Let us choose:  $V_{DS} = \phi_t/2$  and  $i_f = 3$

$$\frac{V_{DS}}{\phi_t} = \sqrt{1 + i_f} - \sqrt{1 + i_r} + \ln \left( \frac{\sqrt{1 + i_f} - 1}{\sqrt{1 + i_r} - 1} \right)$$

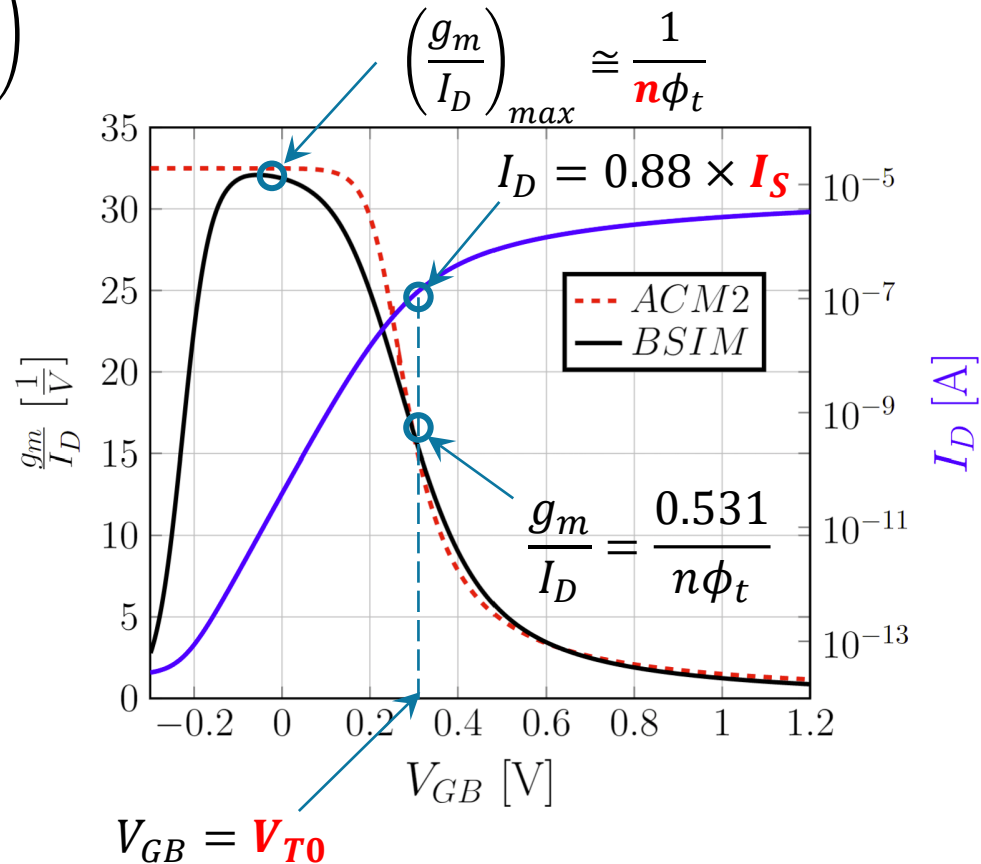
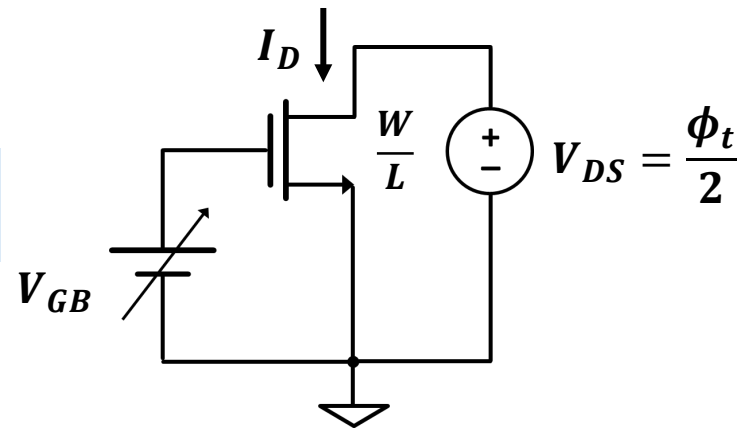
$$\text{Thus: } i_r = 2.12$$

$$\frac{g_m}{I_D} = \frac{d(\ln I_D)}{dV_G} = \frac{2}{n\phi_t(\sqrt{1 + i_f} + \sqrt{1 + i_r})}$$

$$\frac{g_m}{I_D} = \frac{0.531}{n\phi_t} = 0.531 \left( \frac{g_m}{I_D} \right)_{max}$$

$$I_D = I_S(i_f - i_r)$$

$$I_D = (3 - 2.12) I_S = 0.88 I_S$$



# Outline

- **ACM2 : Parameter Extraction**
  - 3-DC-parameter model
    - $V_{T0}$ ,  $I_S$  and  $n$
  - **Transconductances in Saturation**
    - **Sigma & Zeta**
  - DC and small-signals characteristics
  - Example: Inverter CMOS
- **Overview of Open-source environment**
  - Automatic parameter extraction

## ACM2 : Transconductances in saturation

$$I_{Dsat} = \frac{2I_S}{\zeta} q_{dsat} \quad \text{where} \quad \left\{ \begin{array}{l} I_S = \mu C_{ox} n \frac{\phi_t^2}{2} \frac{W}{L} = I_{SH} \frac{W}{L} \\ q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}} \end{array} \right.$$

$$\text{Definitions} \quad \left\{ \begin{array}{l} g_m \triangleq \frac{\partial I_{Dsat}}{\partial V_G} \quad g_d \triangleq \frac{\partial I_{Dsat}}{\partial V_D} \quad g_{msat3} \triangleq \frac{\partial^3 I_{Dsat}}{\partial V_G^3} \end{array} \right.$$

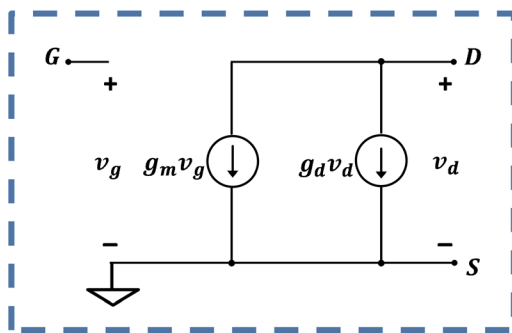
Transconductances in terms of  $q_s$ :

$$g_{msat} = \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \quad g_{dsat} = \sigma \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \quad \Rightarrow \quad g_{dsat} = \sigma g_{msat}$$

$$g_{msat3} = \frac{16I_S}{(n\phi_t)^3} \frac{q_s}{(q_s + 1)^3} \frac{2 - 2\zeta q_s - 3\zeta q_s^2}{(\zeta q_s + 2)^4}$$

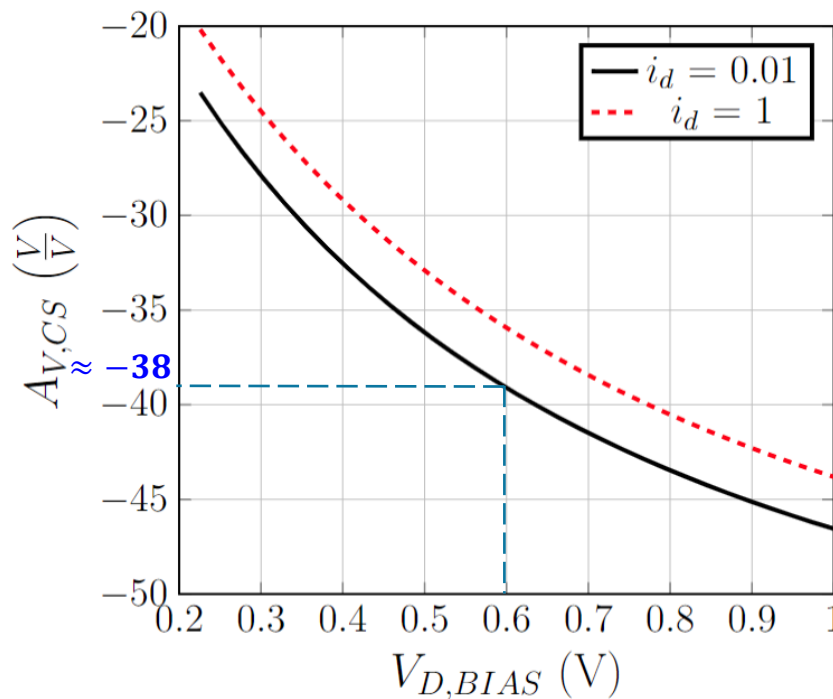
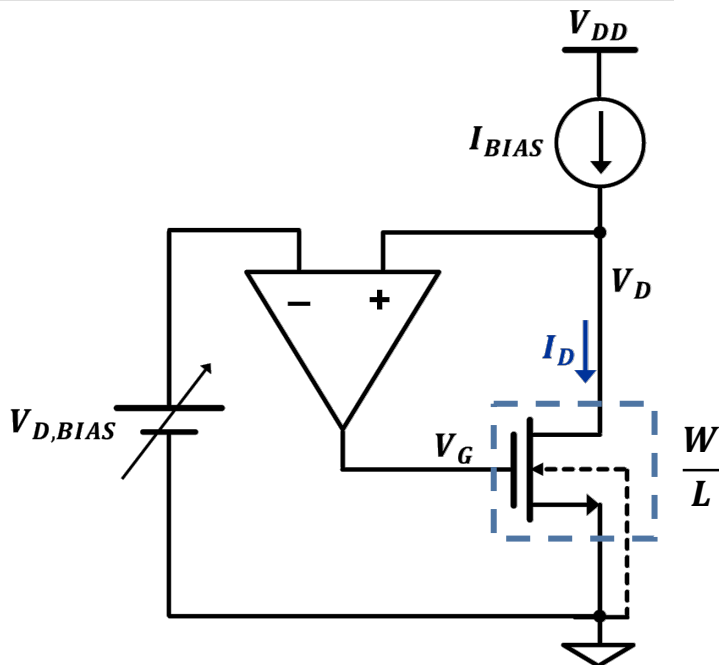
# Extraction of $\sigma$

## Common-Source Intrinsic-Gain method



In saturation :  $g_{dsat} = \sigma g_{msat}$

$$A_{V,CS} = \frac{\Delta v_D}{\Delta v_G} = -\frac{g_{msat}}{g_{dsat}} = \frac{\frac{\partial I_{Dsat}}{\partial V_G}}{\frac{\partial I_{Dsat}}{\partial V_D}} = -\frac{\frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}}{\sigma \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}} = -\frac{1}{\sigma}$$



Example:

$$A_{V,CS} = -\frac{1}{\sigma}$$

$$\sigma = -\frac{1}{(-38)}$$

$$\sigma = 0.026$$

# $\zeta$ extraction

$$i_{dsat} = \frac{2}{\zeta} q_{dsat} \quad q_s = \sqrt{1 + \frac{2}{\zeta} q_{dsat}} - 1 + q_{dsat}$$

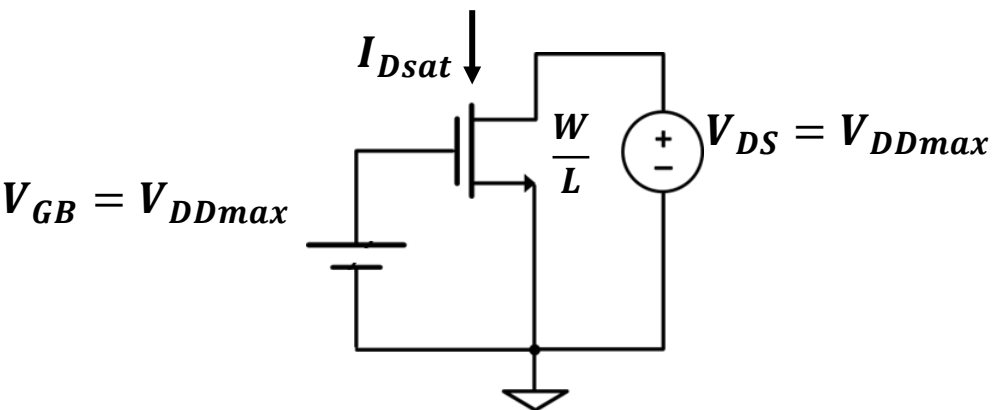
$$\zeta = \frac{2(q_s + 1 - \sqrt{1 + i_{dsat}})}{i_{dsat}}$$

- $q_s$  calculated using parameters ( $V_{T0}$ ,  $n$ ,  $\sigma$ ) and UCCM.

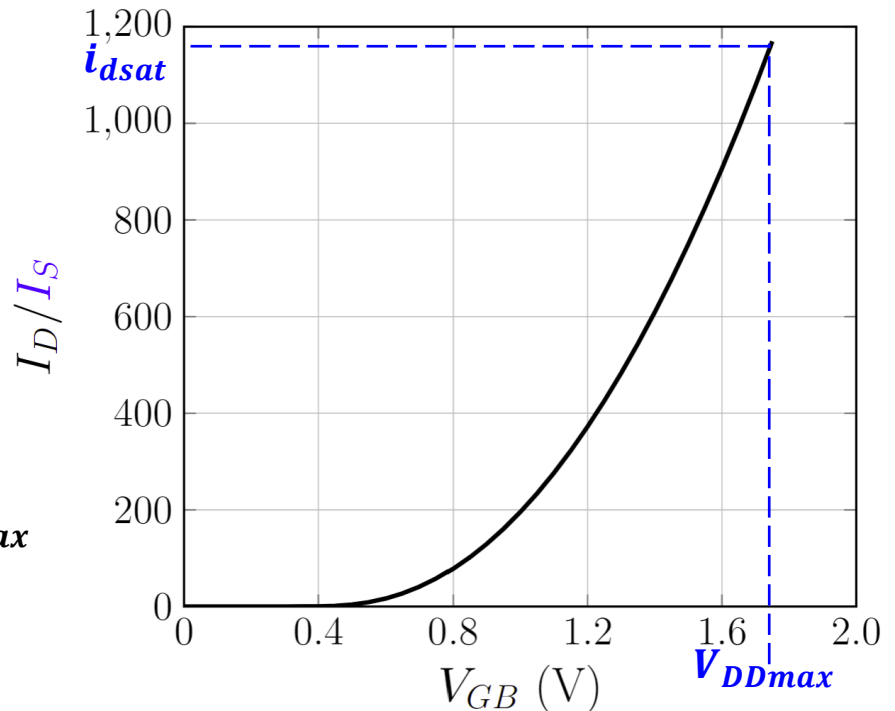


$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

$$\frac{V_P - V_{S(D)B}}{\phi_t} = q_s - 1 + \ln(q_s)$$



- Measure  
 $I_{Dsat} = I_D(V_G = V_D = V_{DDmax} \text{ and } V_S = V_B)$   
 $i_{dsat} = I_{Dsat}/I_S$



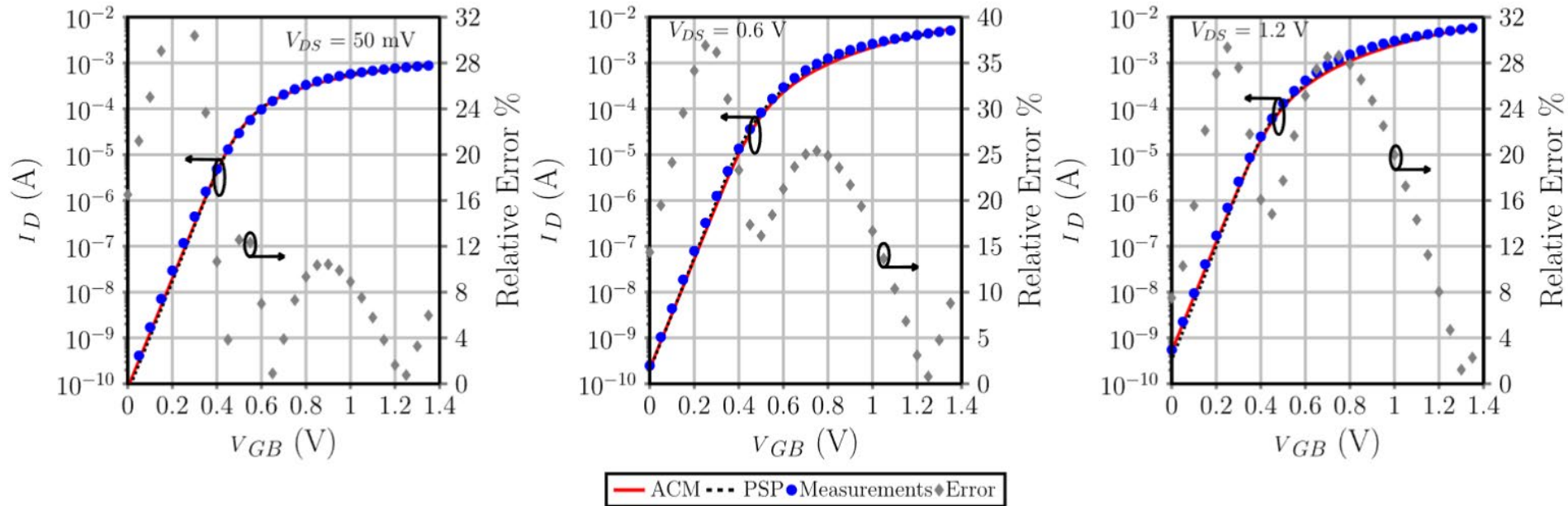


# Outline

- **ACM2 : Parameter Extraction**
  - **3-DC-parameter model**
    - **$V_{T0}$ ,  $I_S$  and  $n$**
  - **Transconductances in Saturation**
    - **Sigma & Zeta**
  - **DC and small-signals characteristics**
  - **Example: Inverter CMOS**
- **Overview of Open-source environment**
  - **Automatic parameter extraction**

# ACM2<sup>1</sup> vs PSP – 130 nm SiGe IHP<sup>2</sup>

## $I_D$ vs $V_{GB}$



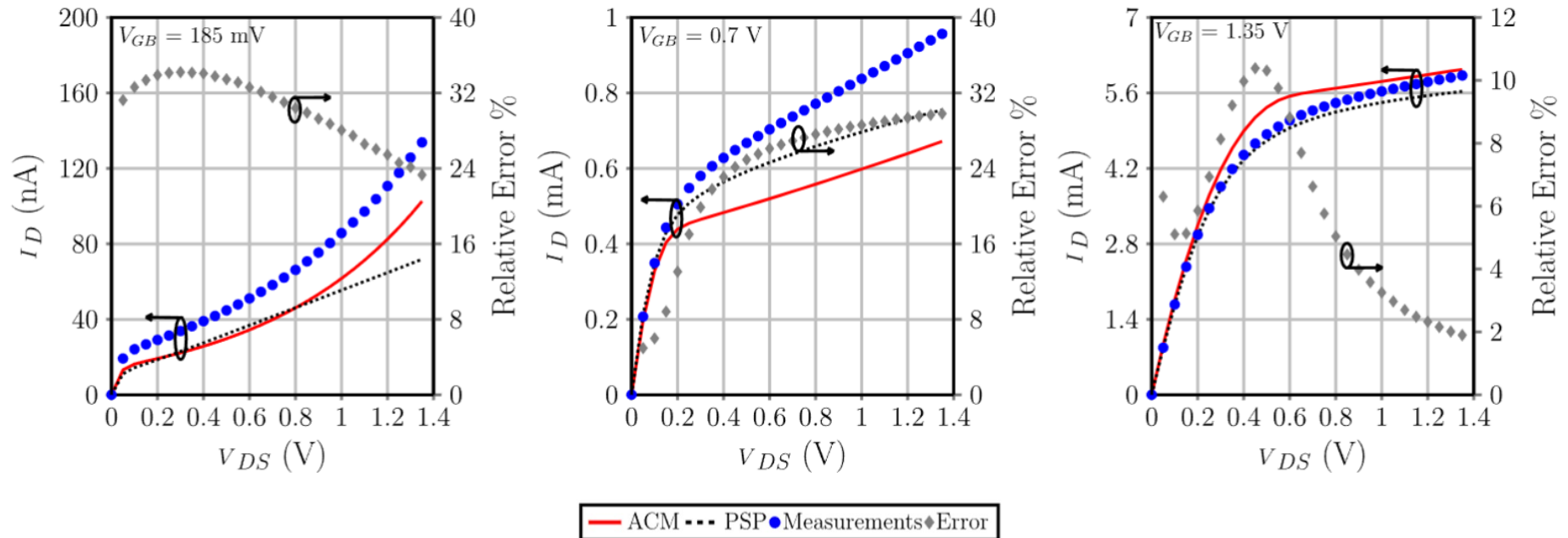
Characteristics of a LVT NMOS bulk transistor with  $W/L = 10\mu\text{m}/120\text{ nm}$ .

<sup>1</sup> ACM2 : implemented in verilog-A, compiled by OPENVAF, simulated in Ngspice

<sup>2</sup> Institut for High-Performance Microelectronics (IHP) open-source PDK

# ACM2<sup>1</sup> vs PSP – 130 nm SiGe IHP<sup>2</sup>

## $I_D$ vs $V_{DS}$

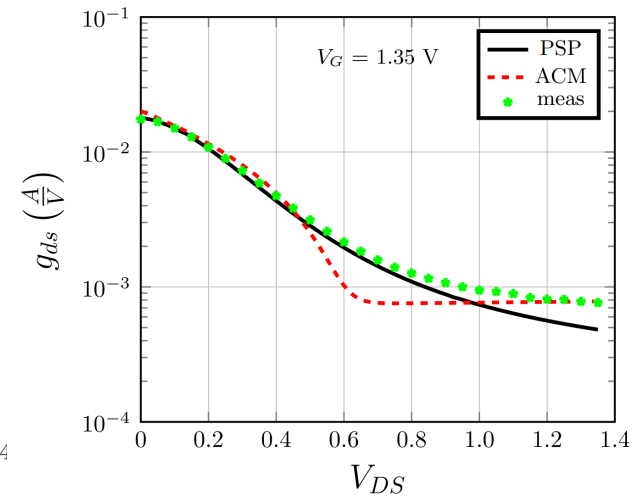
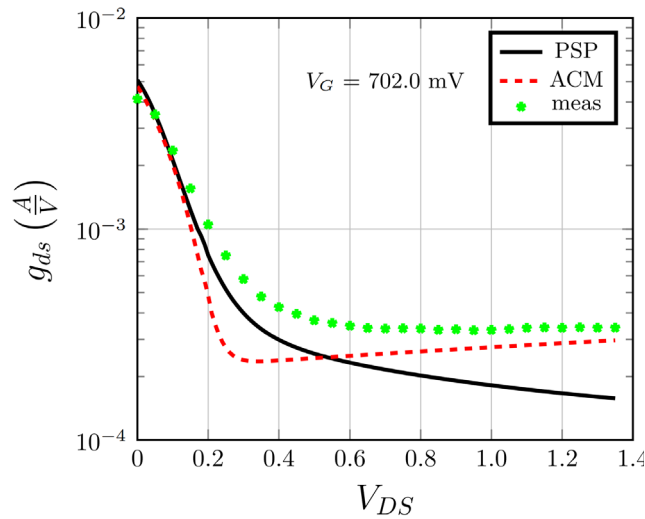
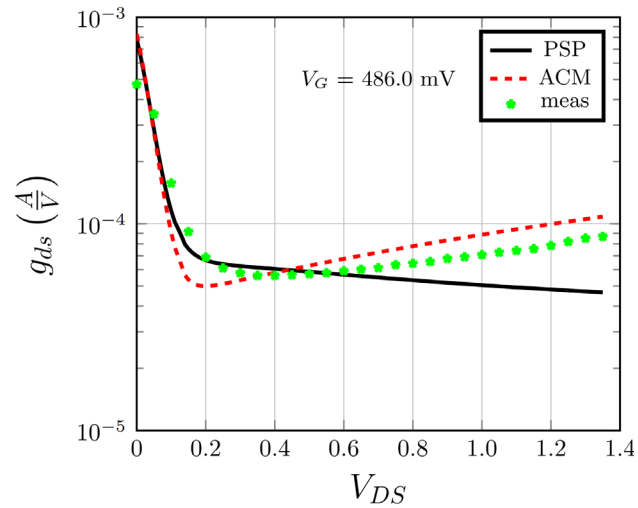
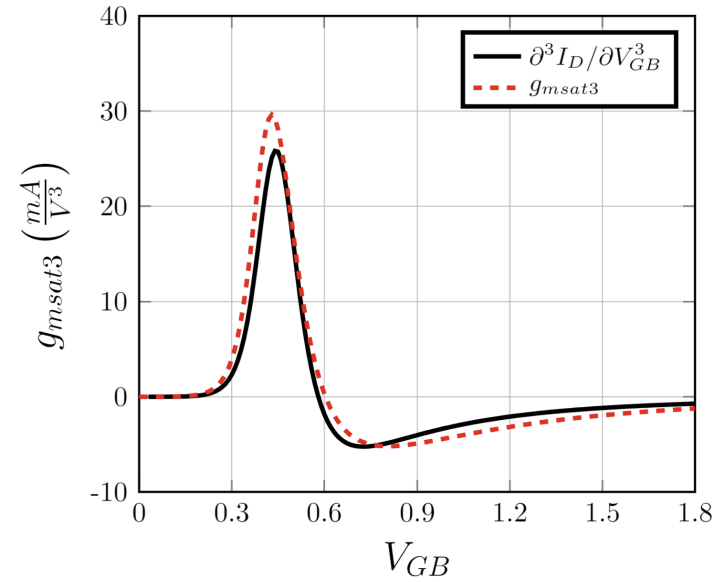
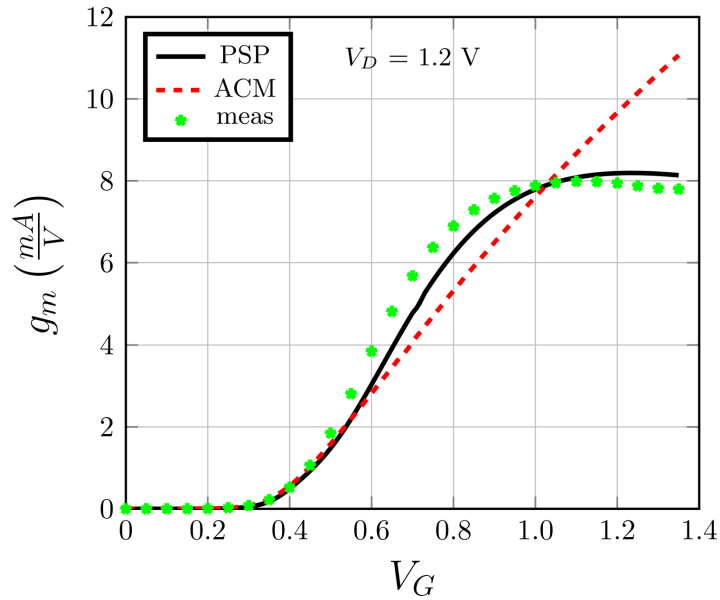


Characteristics of a LVT NMOS bulk transistor with  $W/L = 10\mu\text{m}/120\text{ nm}$ .

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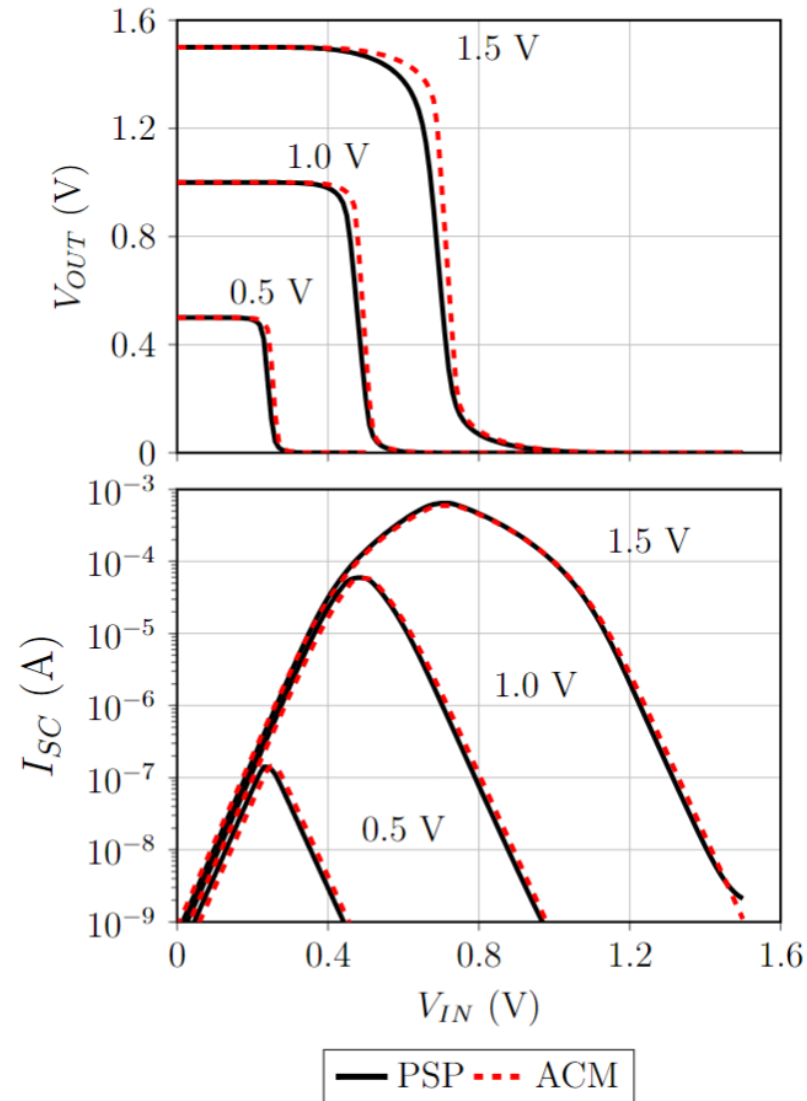
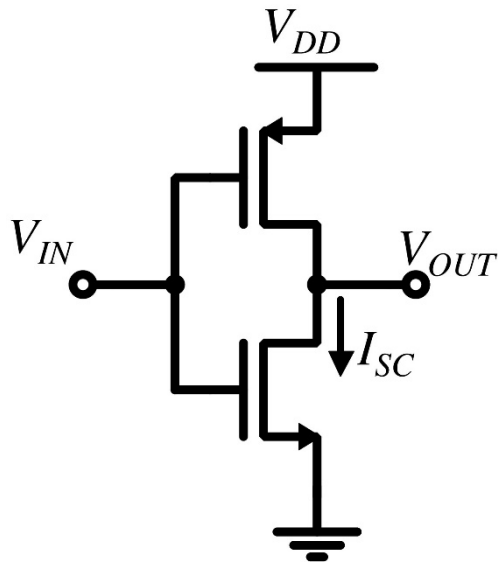
<sup>2</sup> Institut for High-Performance Microelectronics (IHP) open-source PDK

# 5PM-ACM2 : Transconductance gm, gm3 and gds



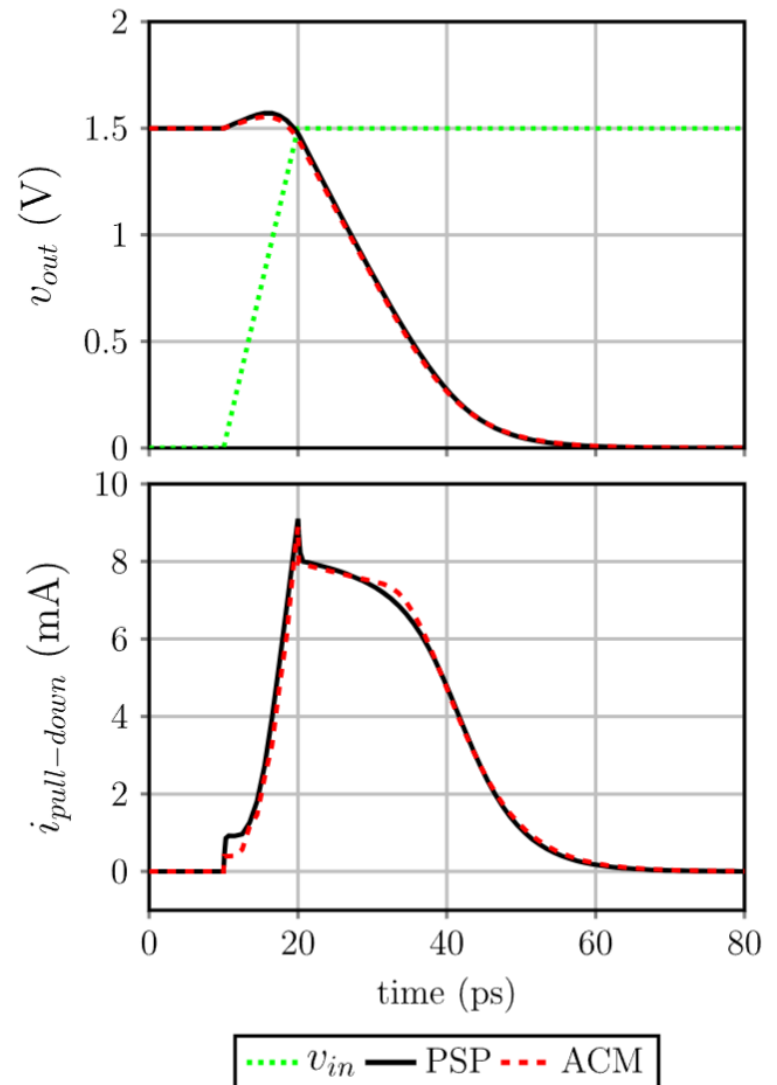
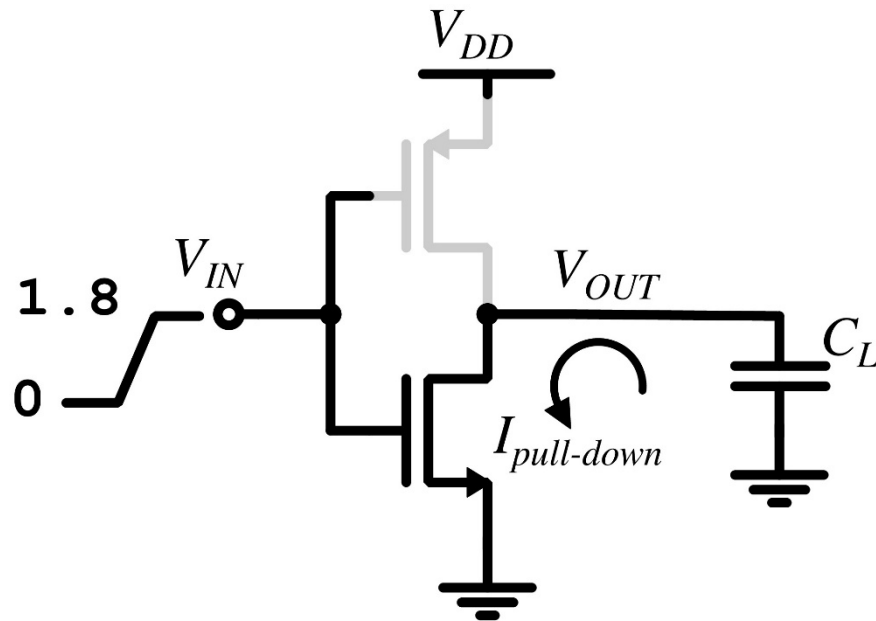
# CMOS Inverter in 130 nm bulk

## VTC and short-circuit current



# CMOS Inverter in 130 nm bulk

## Output Voltage and pull-down current

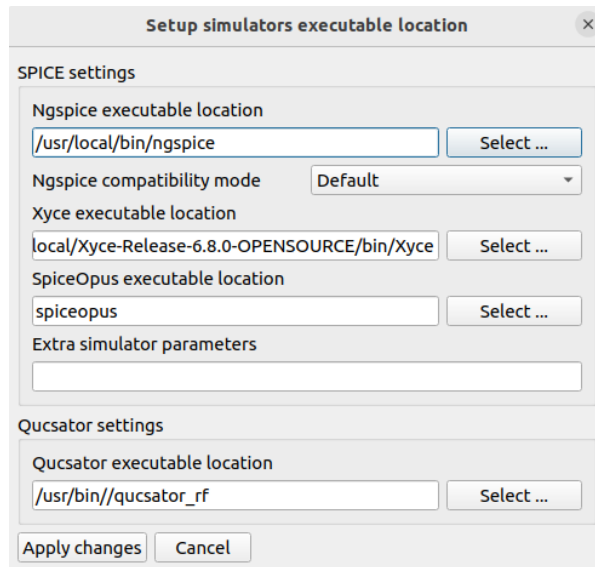


# Outline

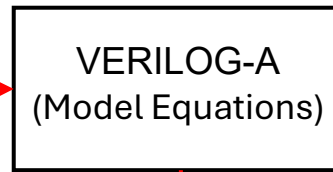
- **ACM2 : Parameter Extraction**
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# Qucs-S: Quite universal circuit simulator with SPICE

## Simulation setup:



SG13G2 PDK





# Qucs-S: Quite universal circuit simulator with SPICE

Device path for the MOS model and parameters (IHP pdk)

.INCLUDE SCRIPT

```
INCLSCR1  
SpiceCode=.LIB /home/acm2/Desktop/IHP-Open-PDK/ihp-sg13g2/libs.tech/ngspice/models/cornerMOSlv.lib mos_tt  
.control  
pre_osdi /home/acm2/QucsWorkspace/psp103_nqs.osdi  
.endc
```

**dc simulation**

DC1

**Parameter sweep**

SW1  
Sim=DC1  
Type=lin  
Param=V1  
Start=0  
Stop=1.5  
Points=301

Input eqs.

Nutmeg

NutmegEq1  
Simulation=SW1  
IDmax=maximum(i(vd))  
IS=11.77e-6  
idsat=IDmax/IS

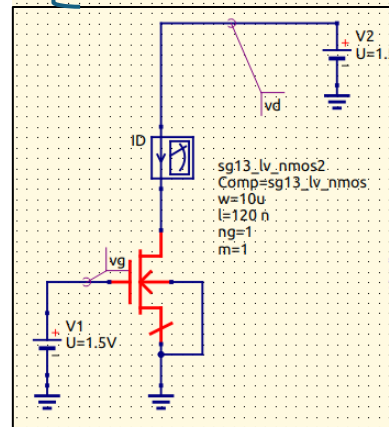
Tabular results

| v(v-sweep) | i(idmax) | idsat |
|------------|----------|-------|
| 0          | 0.00809  | 688   |

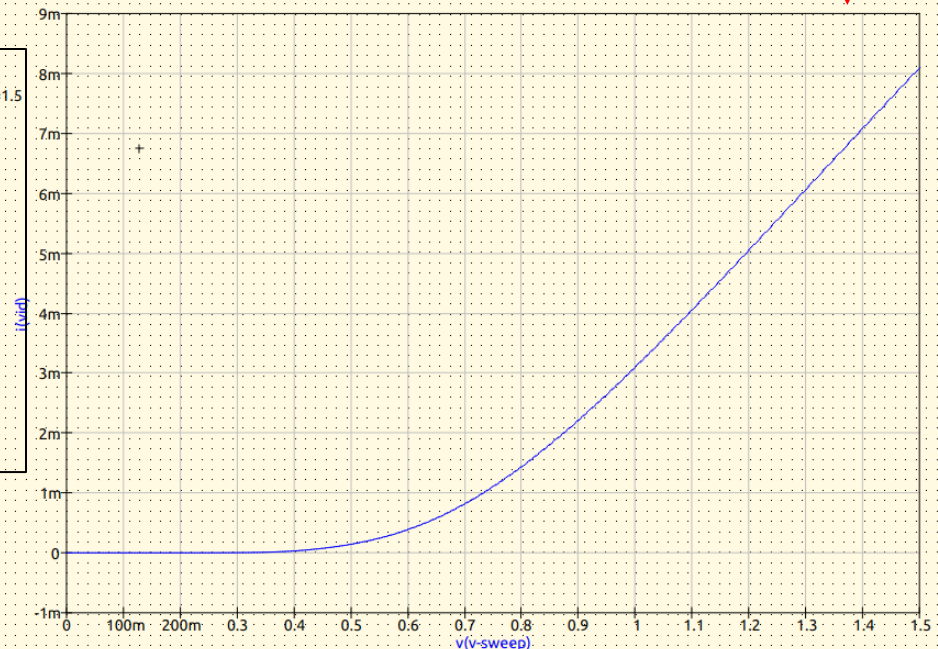
Diagram results

- Ngspice simulations:

- DC
- AC
- TRAN
- S-parameters
- Noise
- Fourier



Circuit  
Schematic



# Github – ACM2

## Github - Content

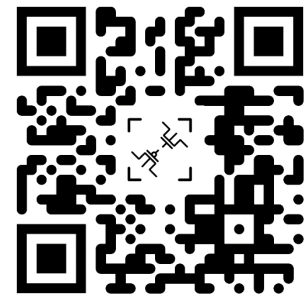
 **ACMmodel** Merge pull request #26 from gabrielmaranhao/main 71ee7cd · 6 months ago

|           |                                                      |
|-----------|------------------------------------------------------|
| Examples  | Update SKY130 and GF180 using ACM examples on xschem |
| Verilog-A | Update PMOS_ACM_2V0.va                               |
| docs      | Delete 5PM_NewCAS.pdf                                |
| LICENSE   | Update LICENSE                                       |
| README.md | Update README.md                                     |

 **README**  ECL-2.0 license

## Advanced Compact MOSFET model (ACM)

ACM is a simple MOSFET model to design and simulate Analog, Mixed-Signal, and RF circuits



Verilog-A code Available!

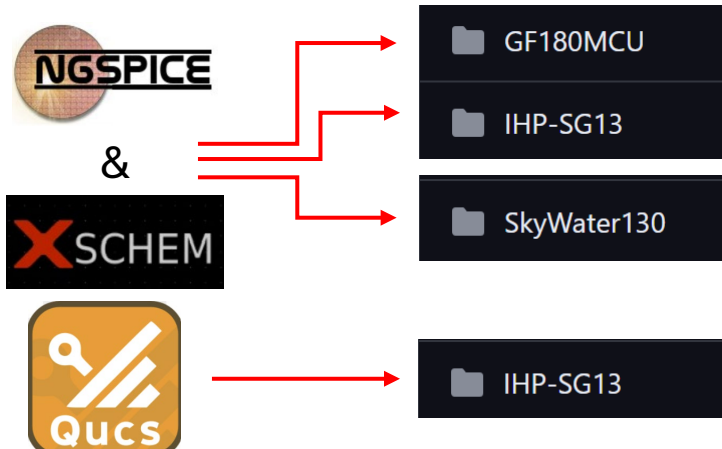
```
MOSFET_model / Verilog-A / NMOS_ACM_2V0.va

ACMmodel Update NMOS_ACM_2V0.va

Code Blame 291 lines (245 loc) · 12.3 KB

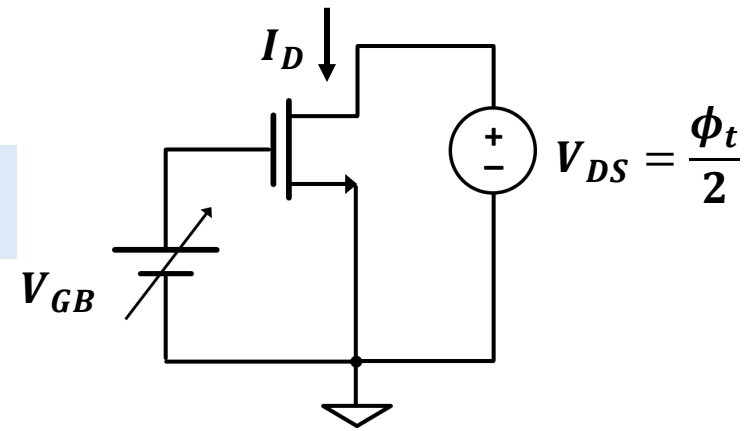
1 //*****
2 // * ACM NMOS model (Verilog-A) *
3 // * 07/2023 V2.0.0 *
4 //*****
5
6 // *****
7 // * Copyright under the ECL-2.0 license *
8 // * Universidade Federal de Santa Catarina *
9 // *
10 // * Current developers: Deni Germano Alves Neto (Doctoral student, UFSC) *
11 // * Cristina Missel Adornes (Doctoral student, UFSC) *
12 // * Gabriel Maranhao (Doctoral student, UFSC) *
13 // *
14 // * Project Supervisors: Prof. Carlos Galup-Montoro *
15 // * Prof. Marcio Cherem Schneider *
16 // *****
17
18 `include "constants.vams"
19 `include "disciplines.vams"
20
21 // function of the algorithm 443 to calculate de normalize charge densities
22 `define algo_443(Z,qn) \
23     if(Z < 0.7385) begin \
24         numeratorD = Z + (4.0/3.0)*Z*Z; \
25         denominatorD = 1.0 + (7.0/3.0)*Z+(5.0/6.0)*Z*Z; \
26         WnD = numeratorD/denominatorD; \
27     end else begin \
28         numeratorD = ln(Z)*ln(Z)+2.0*ln(Z)-3.0; \
29         denominatorD = 7.0*ln(Z)*ln(Z) + 58.0*ln(Z) +127.0; \
30         WnD = ln(Z) - 24.0*(numeratorD/denominatorD); \
```

## Examples of PDKs and circuit simulators using the ACM model



# $I_{S0}, V_{T0}$ and $n$ extraction

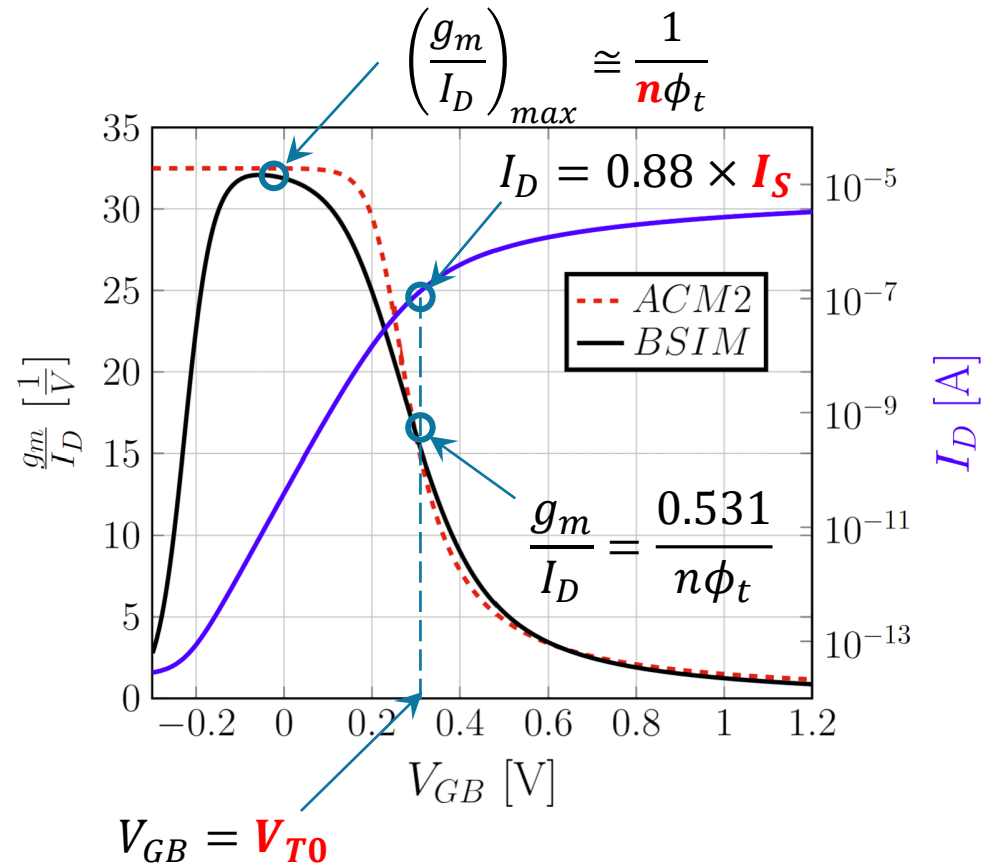
## The $g_m/I_D$ method



We identify the  $\left. \frac{g_m}{I_D} \right|_{max} \rightarrow n$

We calculate  $0.531 \left. \frac{g_m}{I_D} \right|_{max} \rightarrow V_{T0}$

We read the value of  $I_D \rightarrow I_S$



**Practical part  $\rightarrow$  Go to QuCS !**

# Let's go to Qucs-S!

## Useful links:

- Qucs-S oficial page: <https://ra3xdh.github.io/>
- Qucs-S iterative doc: <https://qucs-s-help.readthedocs.io/>
- Ngspice oficial page : <https://ngspice.sourceforge.io/index.html>
- Google Colab: ACM2 & LNA design:  
[https://colab.research.google.com/drive/1s3PKF6pf3zlhTj6jc-qhCLlcGfJ\\_UEE?usp=sharing](https://colab.research.google.com/drive/1s3PKF6pf3zlhTj6jc-qhCLlcGfJ_UEE?usp=sharing)

# Summary – The ACM2 model

- *A truly compact MOSFET model with single-piece functions*
- *Implemented in Verilog-A for simulation*
- *Interchangeable between simulators (SPICE or SPECTRE)*
- *Verify in all three open-source PDKs (Sky130, GF180, IHP-SG13G2)*
- *Helpful to designers (only 5-DC-parameters)*
- *Simplified parameter extraction procedure*
  - *Accepts parameters extracted from simulations or chip measurements*

# Acknowledgments



This project has received funding from the European Union's Horizon Europe research and innovation programme under the HORIZON-KDT-JU-2023-1-IA grant agreement No 101139785

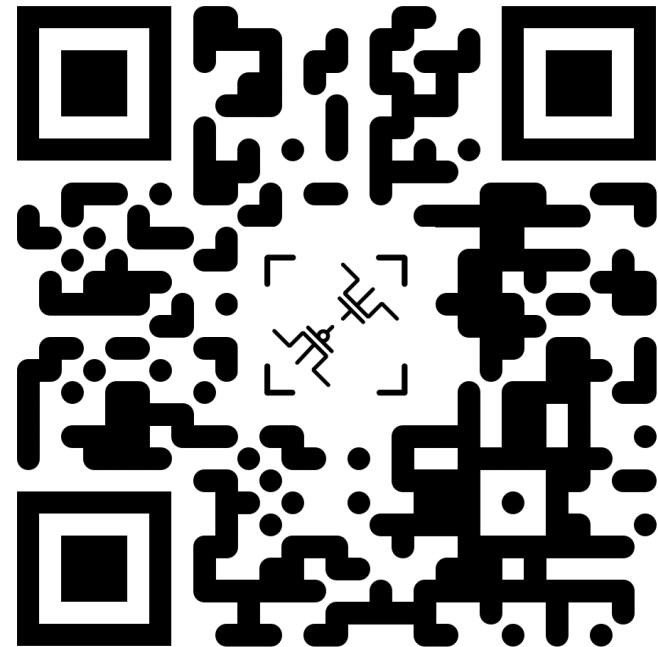


# References

- D. Germano Alves Neto et al., "Design-Oriented Single-Piece 5-DC-Parameter MOSFET Model," in IEEE Access, vol. 12, pp. 87420-87437, 2024.
- ACM2 Github: [https://github.com/ACMmodel/MOSFET\\_model](https://github.com/ACMmodel/MOSFET_model)
- IHP Github : <https://github.com/IHP-GmbH/IHP-Open-PDK>

“Scan me”

- **Available in Github:**
  - DC model
  - Small-signal model
  - Dynamic model
  - Thermal & Flicker noise models (1/f)



# **Step-by-Step extraction**



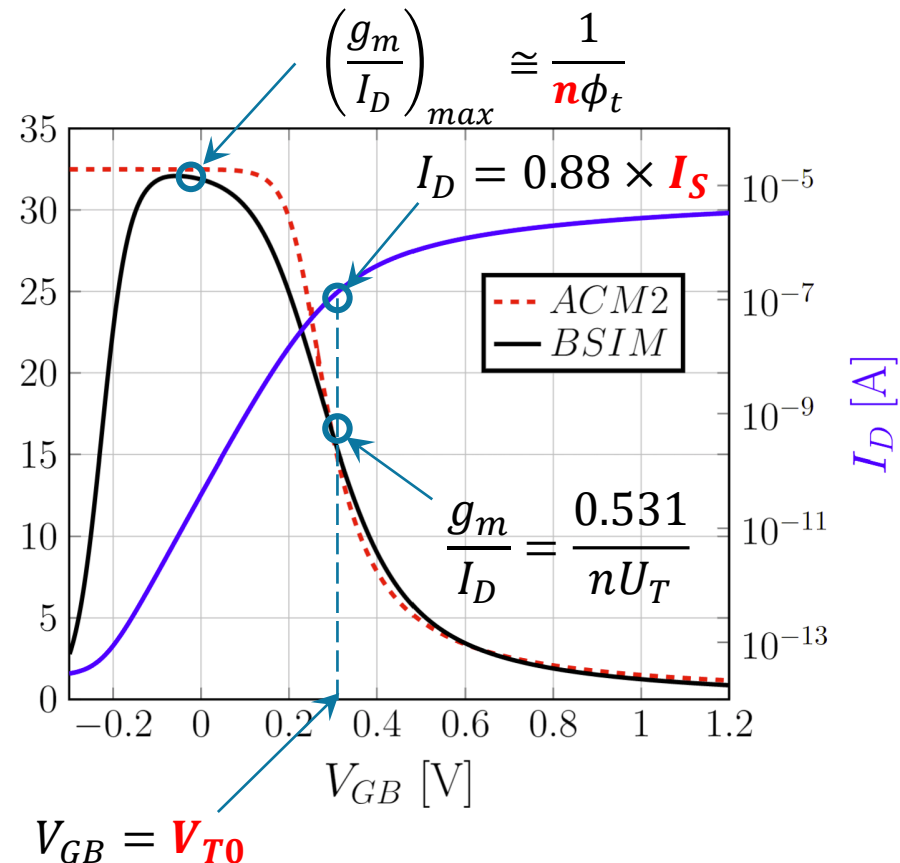
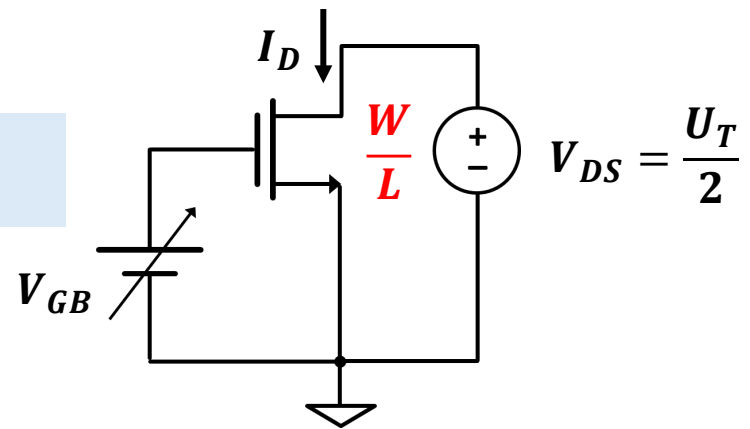
# $I_{S0}, V_{T0}$ and $n$ extraction

## $g_m/I_D$ method – Steps!

1. Assemble the circuit for extraction
2. Define the  $V_{DS}$  value
3. Do a voltage sweep in  $V_{GB}$  and measure the drain current  $I_D$
4. Put on the right side of the plot the  $I_D$  in semilog y axis
5. Plot the  $g_m/I_D$  characteristics using equation in Qucs-s.

$$\frac{g_m}{I_D} = \frac{d(\ln I_D)}{dV_G}$$

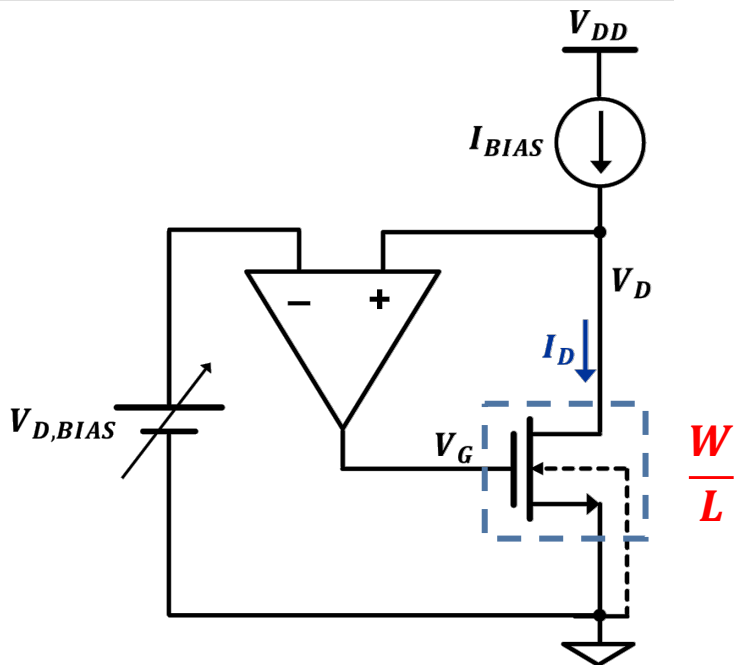
6. With a cursor measure the maximum of  $g_m/I_D$   $\left[\frac{1}{V}\right]$
7. With  $g_m/I_D(\max)$  multiply it by 0.531 to find the point in which  $V_{GB} = V_{T0}$
8.  $I_S$  is the same  $V_{GB}$  value but in the  $I_D$  curve divided by 0.88
9.  $n$  is extracted from the  $g_m/I_D(\max)$  value and  $\phi_t = 25.8$  mV (room temperature)



# $\sigma$ extraction

## *Common-Source Intrinsic-Gain method - Steps!*

$$A_{V,CS} = \frac{\Delta v_D}{\Delta v_G} = -\frac{g_{msat}}{g_{dsat}} = -\frac{1}{\sigma}$$



1. Assemble the circuit for extraction
2. Define the  $I_{BIAS} = 100$  nA (weak inversion)
3. Use Qucs-S equations to calculate the derivate of  $v_d$  and  $v_g$
4. Using equations set the equation for sigma
5. Do a voltage sweep in  $V_{D,BIAS}$  from 0.2 V to 1 V
6. With a cursor measure the value for sigma at  $V_D = V_{DDmax}/3$

# $\zeta$ extraction

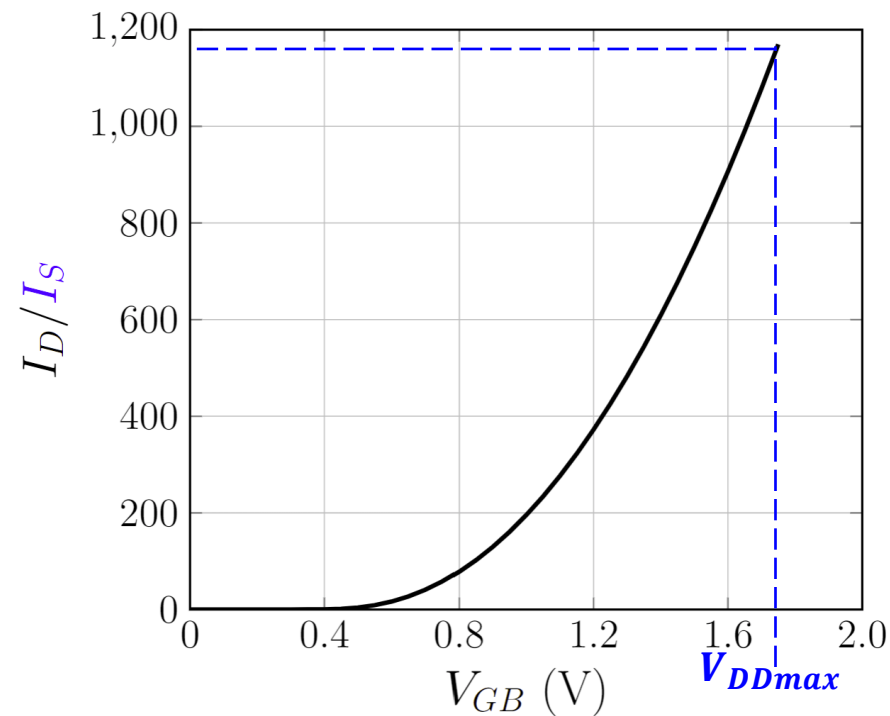
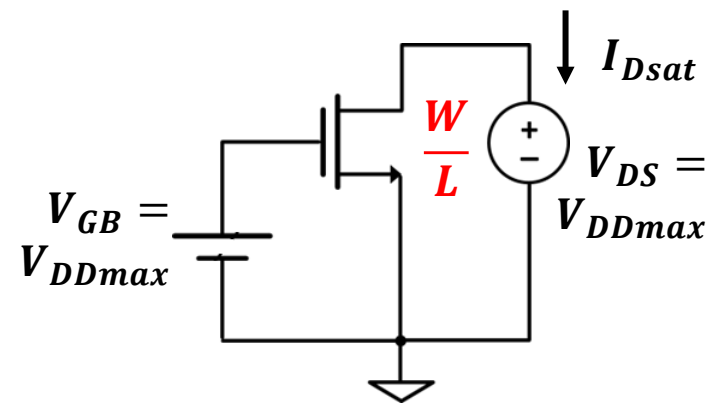
## Saturation method – Steps !

1. Assemble the circuit for extraction
2. Define  $V_G = V_D = V_{DDmax}$  and  $V_S = V_B = 0\text{ V}$
3. Using equations set the equation for  $i_{dsat} = I_{Dsat}/I_{S0}$
4. Insert a table in Qucs-s to evaluate  $i_{dsat}$

Google Colab: ACM2 & LNA design:

[https://colab.research.google.com/drive/1s3PKF6pf3zIhITj6jc-qhCLlcGfJ\\_UEE?usp=sharing](https://colab.research.google.com/drive/1s3PKF6pf3zIhITj6jc-qhCLlcGfJ_UEE?usp=sharing)

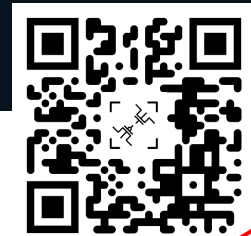
5. Using the Colab, in the section Zeta extraction, we will insert the value of the appropriate  $i_{dsat}$  for the python code which calculate qs transcendant equation and then zeta!





## Part 4

# Design Methodology-Application to Low Noise Amplifier



Sylvain Bourdel



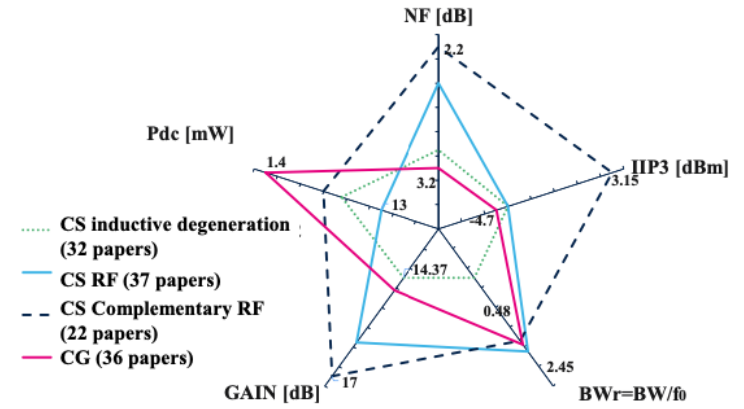
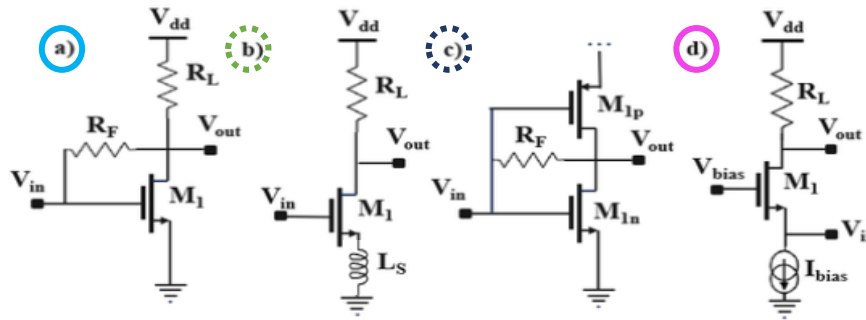
[https://github.com/ACMmodel/MOSFET\\_model](https://github.com/ACMmodel/MOSFET_model)

[https://colab.research.google.com/drive/1s3PKF6pf3zIhITj6jc-ghCLlcGfJ\\_UEE?usp=sharing#scrollTo=EMGo7aUzyukW](https://colab.research.google.com/drive/1s3PKF6pf3zIhITj6jc-ghCLlcGfJ_UEE?usp=sharing#scrollTo=EMGo7aUzyukW)

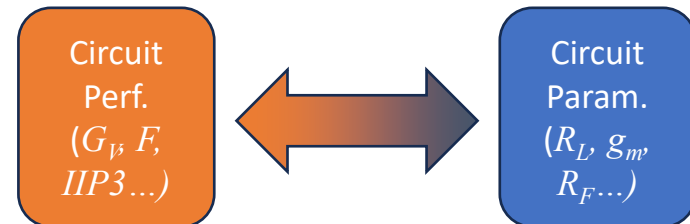
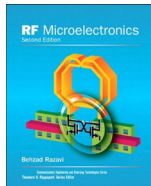
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## • 1<sup>st</sup> Step: Architecture Choice



## • 2<sup>nd</sup> Step: Circuit Analysis

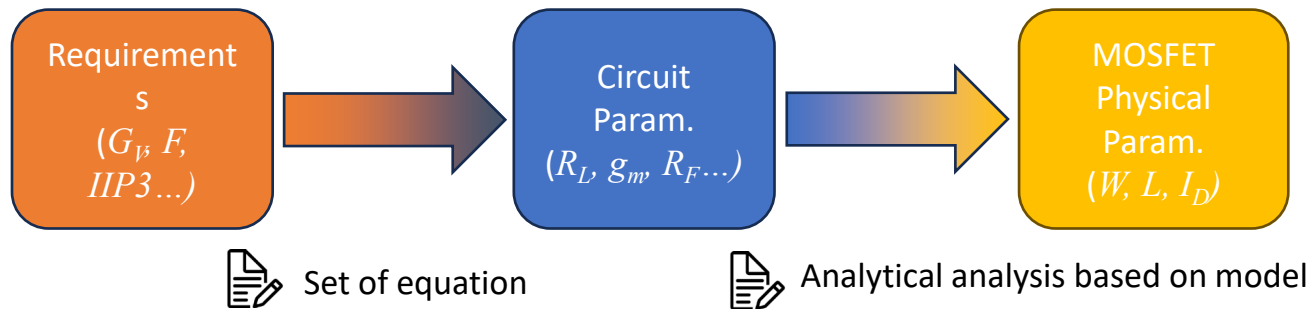


$$|G_T| = |G_v|Q_{IN} = \frac{(G_m R_F - 1)R_O}{(R_O + R_F)} \sqrt{1 + Q_P^2}$$

# Overview of Design Methods for RFIC

## General Principle

- 3<sup>rd</sup> Step: Circuit Sizing

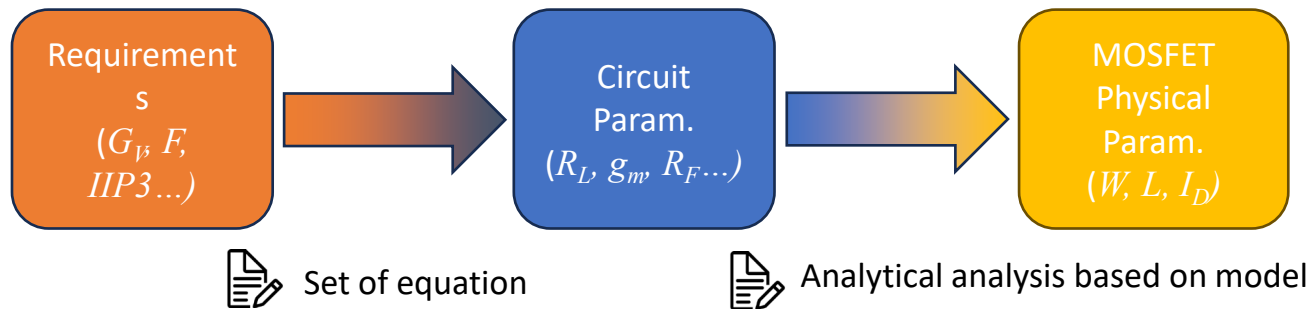




# Overview of Design Methods for RFIC

## General Principle

- 3<sup>rd</sup> Step: Circuit Sizing



### ***Region based model***

$$sat. \quad I_d = 2.K_n \frac{W}{L} \left[ (V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

$$lin. \quad I_d = K_n \frac{W}{L} (V_{gs} - V_t)^2 \left( 1 + \frac{k_{en}}{L} V_{ds} \right)$$

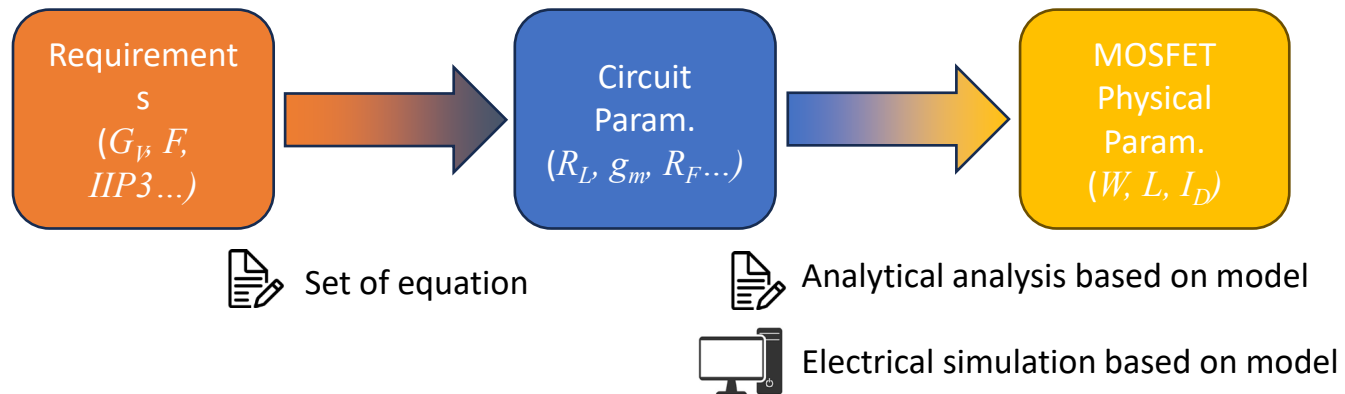
PROS: simple

CONS: Inaccurate with short channel MOS

# Overview of Design Methods for RFIC

## General Principle

- 3<sup>rd</sup> Step: Circuit Sizing



### **BSIM3 / UTSOI compact model**

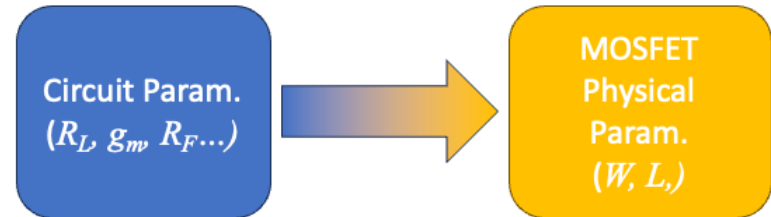
#### *PROS:*

- Accurate
- Direct relationship between requirements and physical parameters

#### *CONS:*

- Increases the gap between Physic and design
- Long optimization

- **Issue in the sizing step**



- How to maintain simplicity **and** accuracy with the scaling and especially Short Channel Effects (SCE) at the sizing step. ?

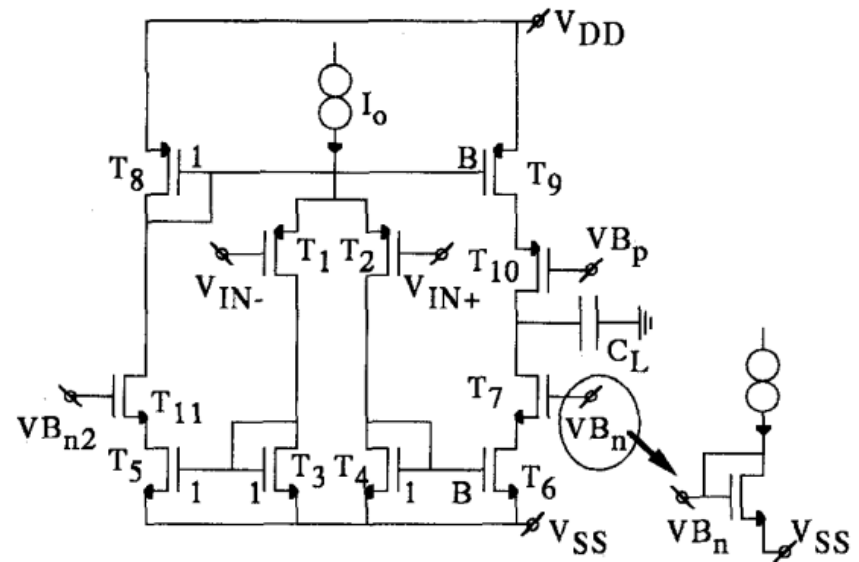
- **A possible solution**

- $g_m/I_d$  design approaches
  - LUT based
  - ACM or EKV based

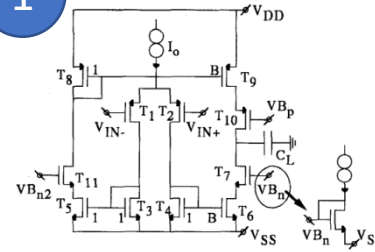
[Silveira, Flandre, Jespers – JSSC 1996]

## General Principle

- For a given structure

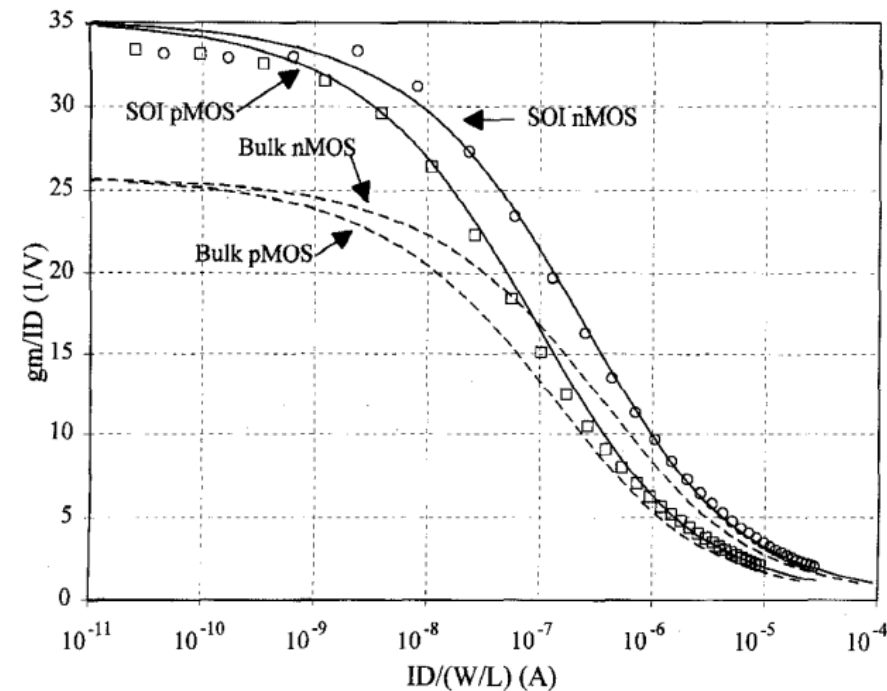


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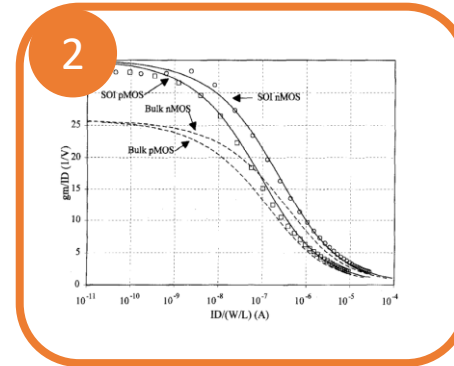
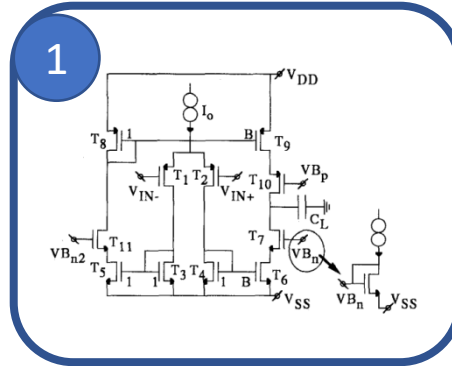


### General Principle

- For a given structure
- Methods are based on
  - ⇒ Extraction of MOS parameters (LUT)
    - $I_D/(W/L)$  vs  $g_m/I_D$
    - $g_m/I_D$  vs  $V_{gs0}$



### General Principle



- For a given structure
- Methods are based on
  - ⇒ Extraction of MOS parameters (LUT)
    - $I_D/(W/L)$  vs  $g_m/I_D$
    - $g_m/I_D$  vs  $V_{gs0}$

⇒ Set of equations

$$SR = \frac{B \cdot I_{D1}}{C_L}$$

$$f_T = \frac{B \cdot g_{m1}}{2\pi \cdot C_L} \quad (5)$$

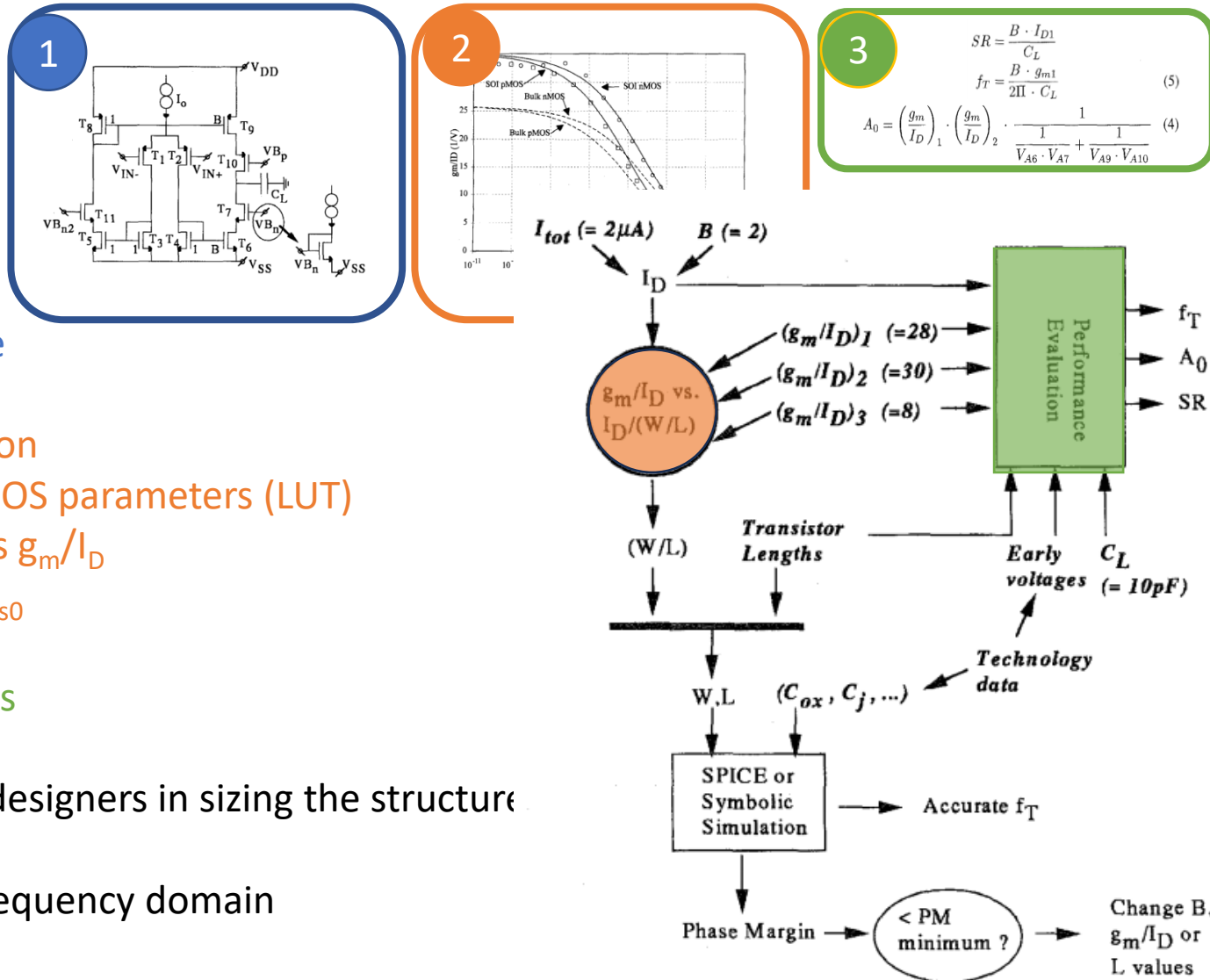
$$A_0 = \left( \frac{g_m}{I_D} \right)_1 \cdot \left( \frac{g_m}{I_D} \right)_2 \cdot \frac{1}{\frac{1}{V_{A6} \cdot V_{A7}} + \frac{1}{V_{A9} \cdot V_{A10}}} \quad (4)$$

# Overview of Design Methods for RFIC

## $g_m/I_D$ approach based on LUT

### General Principle

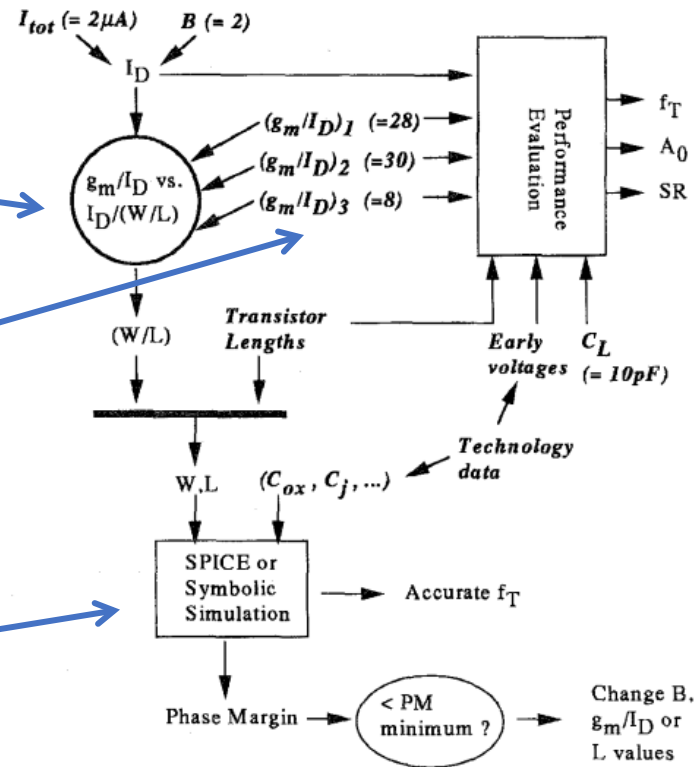
- For a given structure
- Methods are based on
  - Extraction of MOS parameters (LUT)
    - $I_D/(W/L)$  vs  $g_m/I_D$
    - $g_m/I_D$  vs  $V_{gs0}$
- Set of equations
- Methods helps the designers in sizing the structure
- Quite used in low frequency domain



### LIMITATIONS ?

- Time Consuming
- Lack of precision (parametric set of characteristics)
- Long optimization sequence

Need for a model based approach





### All Region 3PM model

- EKV – [Enz, Krummenacher, Vittoz]
- ACM – [Schneider, Galup]



- With a 3PM model we have :

$$\frac{g_m}{I_D} = \frac{2}{n\phi_t \left(1 + \sqrt{1 + i_f}\right)} \quad g_m = \frac{2I_S}{\phi_t} \left(\sqrt{1 + i_f} - 1\right) \quad V_P - V_{S(D)} = \phi_t \left[ \sqrt{1 + i_{f(r)}} - 2 + \ln \left( \sqrt{1 + i_{f(r)}} - 1 \right) \right]$$

$$V_P \cong \frac{V_G - V_{T0}}{n}$$

- PROS: The sizing is straightforward
- CONS: Inaccurate with SCE

Non-linearities can't be captured  
gds effect can't be captured

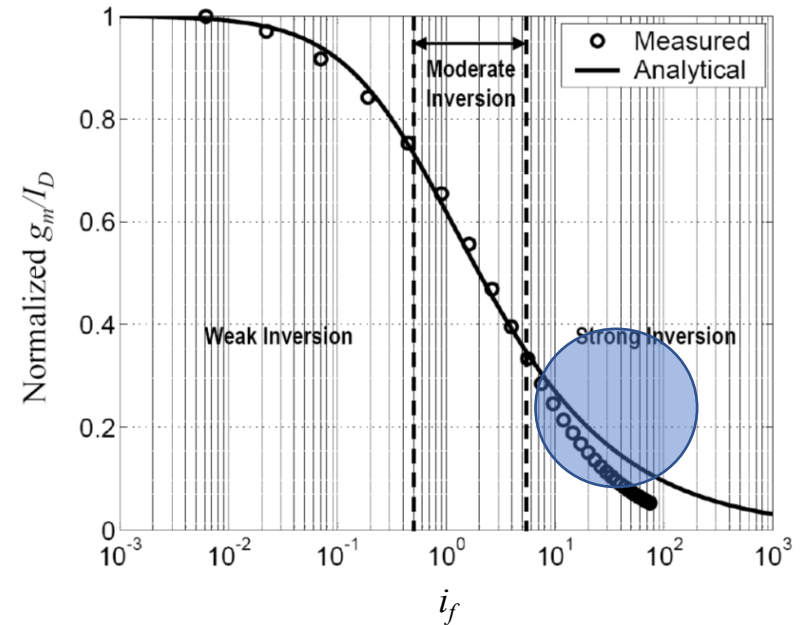
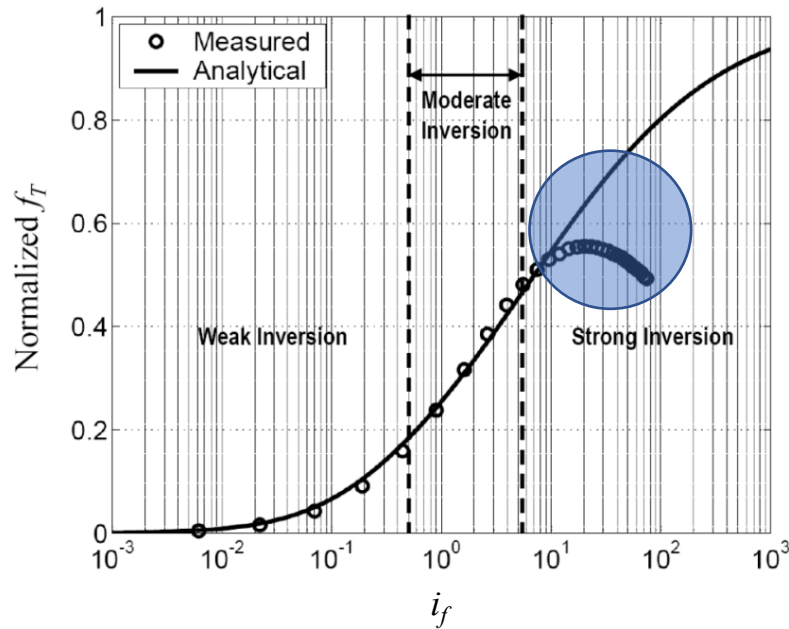


Can we design RFICs ?

### Design Regions

- $f_t$  grows with  $i_f$

 Year 2000 (RF-180nm)



### Design Regions

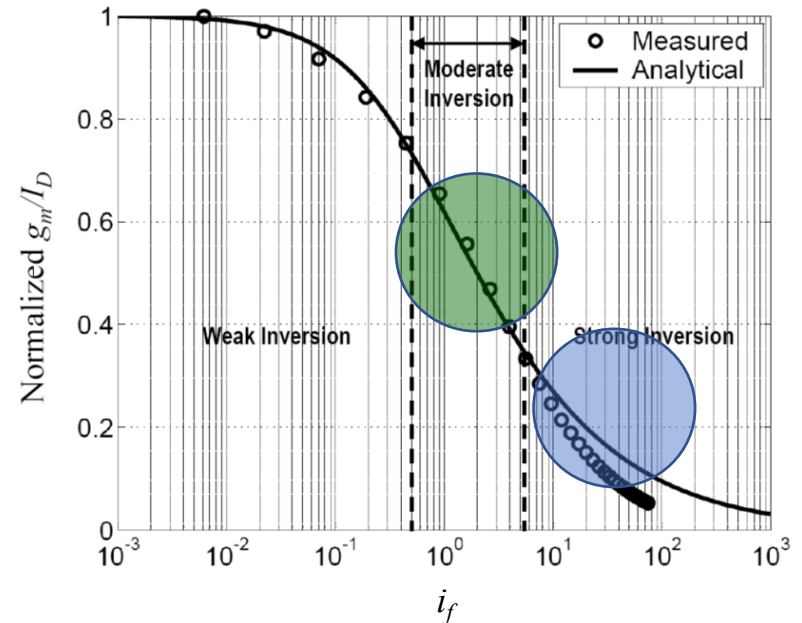
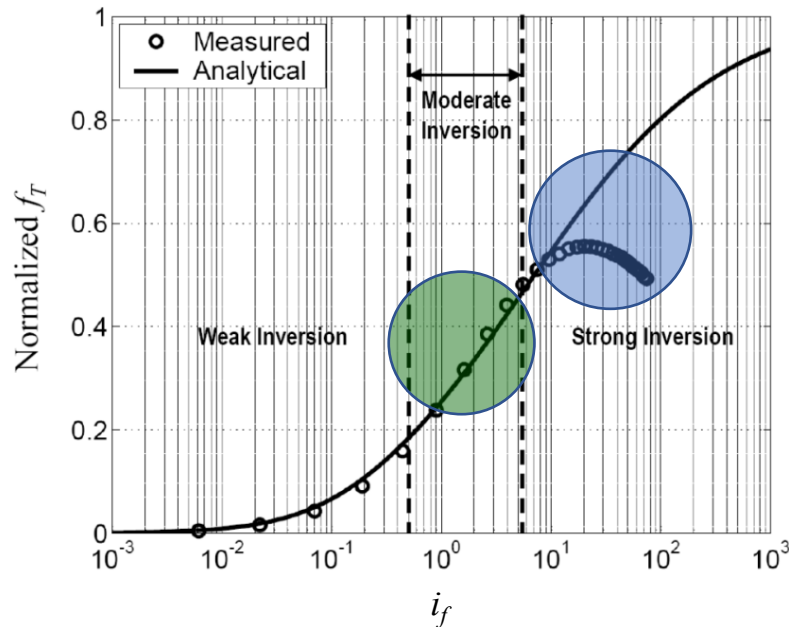
- $f_t$  grows with  $i_f$  but  $g_m/I_D$  reduces with  $i_f$



Year 2000 (RF-180nm)



Year 2016 (RF-22nm) up to now



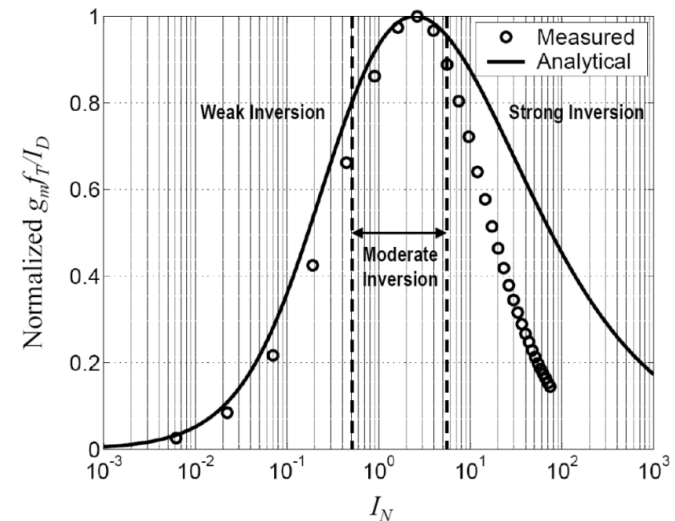
### $g_m f_T / I_D$ : A First Approach

A. Shameli et P. Heydari, « Ultra-Low Power RFIC Design Using Moderately Inverted MOSFETs: An Analytical/Experimental Study », *RFIC*, 2004.

- For RF design
- A FOM that maximize the gain bandwidth product

$$\frac{g_m f_T}{I_D}$$

- Gives the  $i_f$  that produces the best GBW product
- Not optimal :
  - the gain or the bandwidth might be oversized.
  - Do not depends on the topology

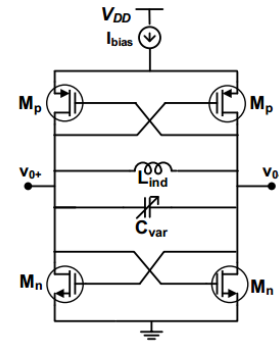
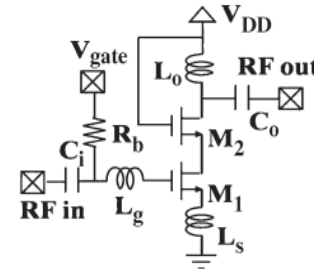


LNA

LO

### Function Based FoM

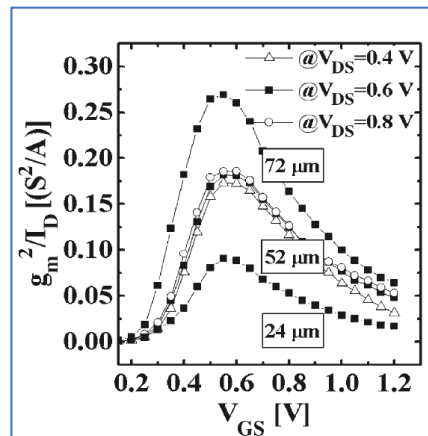
- For a given Function
- Define a FoM for the function



$$\text{FoM}_{\text{LNA}} = \frac{G}{(F-1) \cdot P} \propto \frac{g_m^2}{\left(\frac{I_D}{g_m}\right) \cdot I_D} \propto \left(\frac{g_m^2}{I_D}\right)^2$$

- Find the  $i_f$  that maximizes the FoM

I. Song et B.-G. Park, « A Simple Figure of Merit of RF MOSFET for Low-Noise Amplifier Design », *Electron Device Lett. IEEE*, vol. 29(12), 2008.



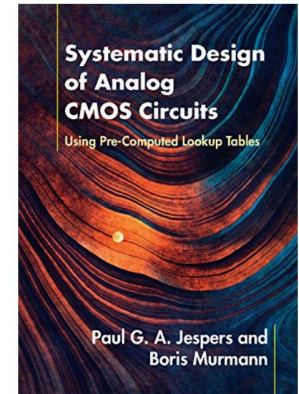
$$PN(\Delta f) = 10 \log \left( \frac{k_B T \pi^2}{64 Q^2} \frac{g_m}{I_D} \frac{1}{I_D} \frac{f_0}{(\Delta f)^2} \lambda \right)$$

R. Fiorelli ; J. Núñez ; F. Silveira; “All-Rnversion region  $g_m/I_D$  methodology for RF circuits in FinFET technologies”  
2018 *NEWCAS*

### Circuit Based Method

#### Circuit Sizing Based on LUT

- For a given Circuit
- Similar to the one introduced by Silveira, Flandre, Jespers in 1996
- Capacitance and noise effects can be captured in LUT for bandwidth and noise analysis
- Well known and well referenced
- But ...
  - Still complicated
  - The gap between physics and design remains
  - No analytical relationships to size « by hand »



#### Circuit Sizing Based on ACM

- Equations can be derived for a given circuit
- Quasi analytical approach
  - R-F LNA designed with 3PM-ACM [Bourdel - ICECS 2019]
- Gds and gm3 (NL) are taken into account
  - R-F LNA designed with 7PM-ACM [Bouchoucha – ISCAS 2023]
  - CG LNA designed with 5PM-ACM2[Alves Neto – ACCESS 2024]



### Resistive Feedback LNA design using a 7-parameter design-oriented model for advanced technologies

Mohamed Khalil Bouchoucha<sup>1,2</sup>, Dayana A. Pinc

Manuel J. Barragan

<sup>1</sup>STMicroelectronics, 38920 Crolles, France, <sup>2</sup>TiMA I

ANALOG CATHELIN

AND CARLOS GALUP-P

<sup>1</sup>Department of Electrical and Electroni

<sup>2</sup>STMicroelectronics, 38920 Crolles, Fr

<sup>3</sup>Univ. Grenoble Alpes, CNRS, Grenob

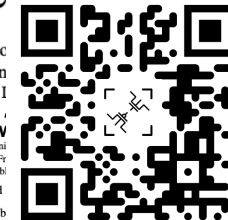
Corresponding author: Mohamed

This work was supported in part b

Nacional de Desenvolvimento Cie

STMicroelectronics, Crolles, France

https://github.com/ACMmodel/MOSFET\_model



lippe Cathelin<sup>1</sup>, Jean-Michel Fournier<sup>2</sup>,

Bourdel<sup>2</sup>

<sup>1</sup>Grenoble Alpes, 38000 Grenoble, France

BOURDEL, JEAN-MICHEL FOURNIER, ILLI,

E)

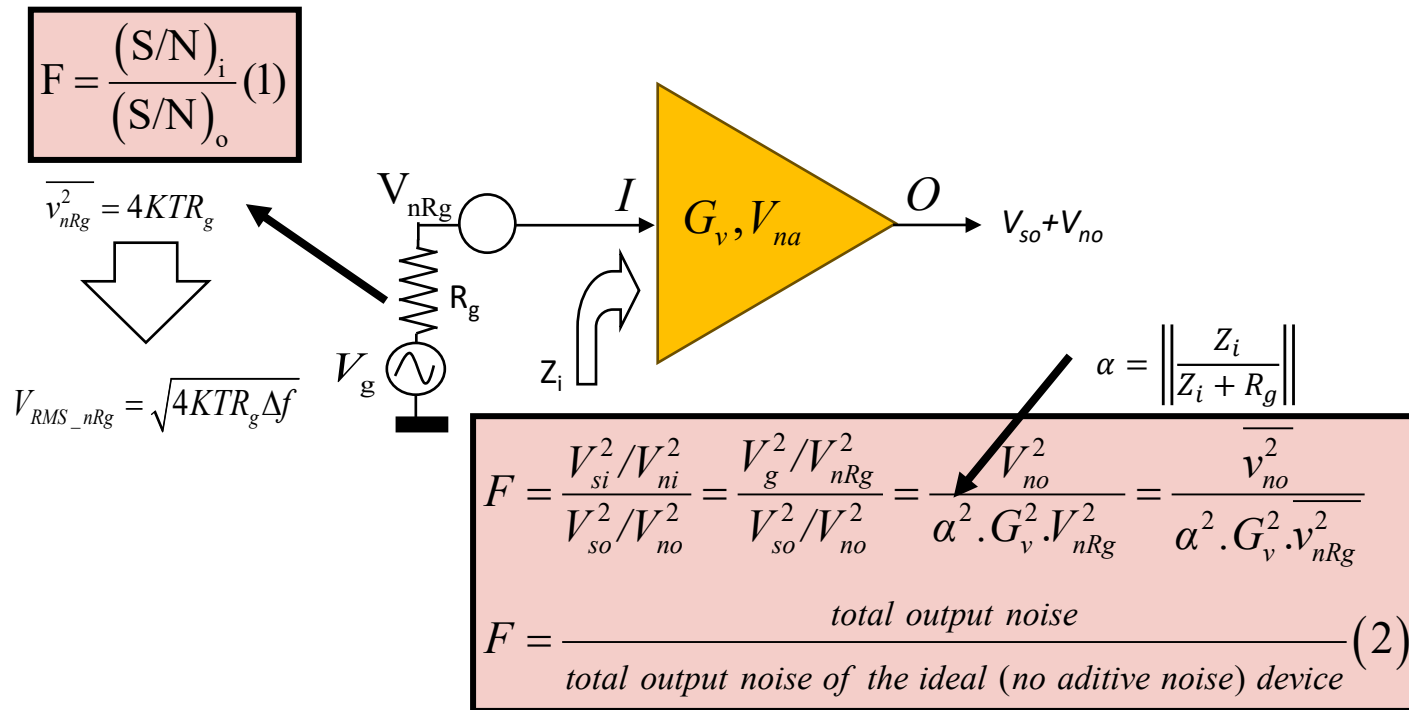
anópolis 88040-900, Brazil

de Nivel Superior, Brazil; in part by the Conselho

Laboratory, Grenoble, France; and in part by

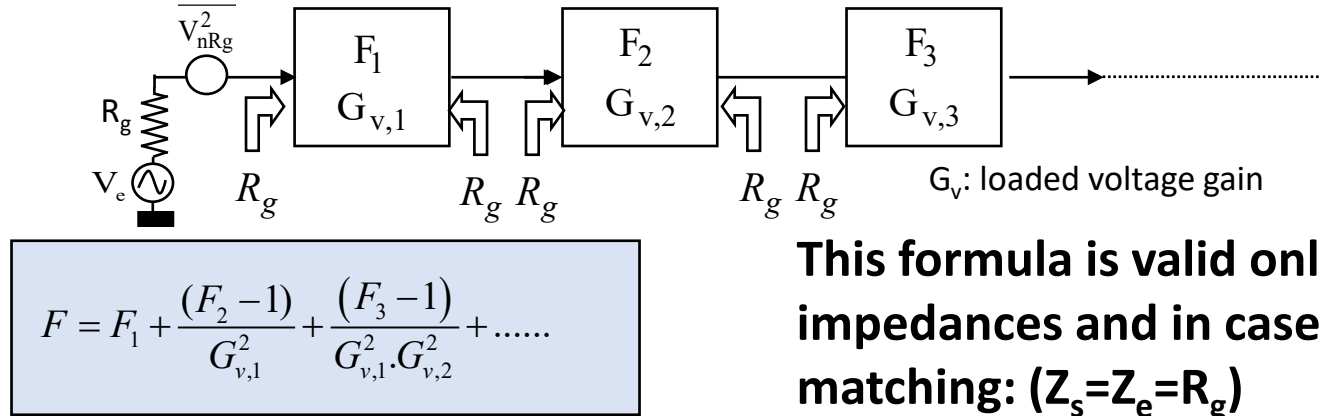
- Overview of Design Methods for RFIC
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- Inversion Level Based Method for R-Feedback LNA with ACM-2

- NF is the signal to noise ratio (SNR) degradation (Eq. 1)



- NF also express the quantity of noise added by the stage regarding the noise delivered by the source ( $R_g$ ) on  $\Delta f$ . (Eq. 2)





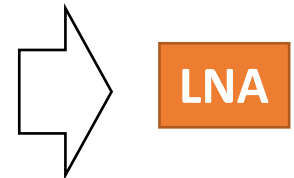
***In RFIC, impedances are complex and never equal. This formula cannot be used as is. However, the following statements remain valid in RFIC.***

The gain of the first stages reduces the NF of the following ones.

=> The receiver chain must start by amplifiers

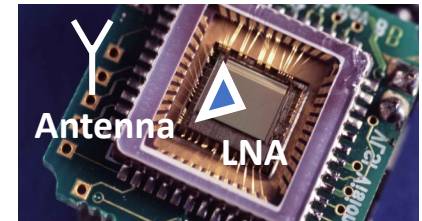
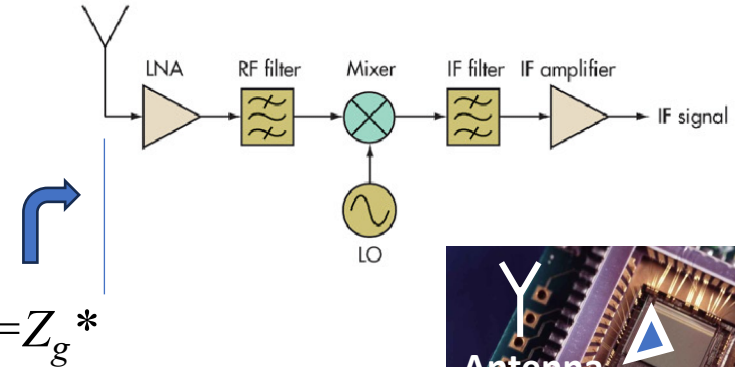
=> The NF of the first stage must be as low as possible

Note that lossy stages ( $G_v < 1$ ) increase the NF, especially when they are in front of the receiver chain (***antenna filter, image rejection filter, antenna switch*** ).



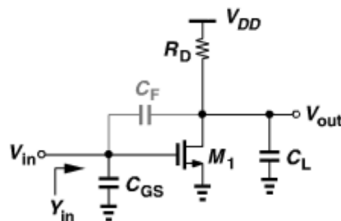
### System point of vue

- LNA is the first device due to Friss formula
- Antenna is mainly 50  $\Omega$ .



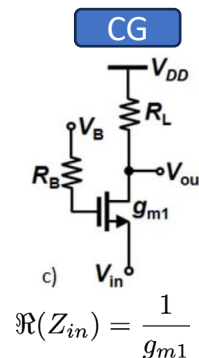
### In CMOS

- Generally, the input impedance is capacitive  $\Re(Y_{11})$  is very low
- For exemple : CS amplifier

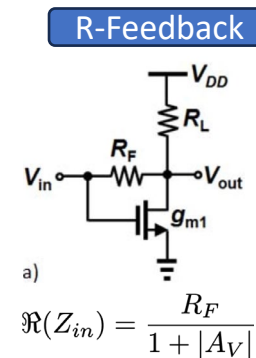


$$\Re(Y_{in}) = R_D \parallel 1/g_m$$

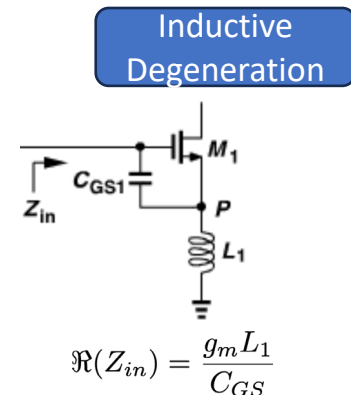
$$\Im(Y_{in}) = \frac{1}{C_{GS}C_F\omega}$$



$$\Re(Z_{in}) = \frac{1}{g_{m1}}$$



$$\Re(Z_{in}) = \frac{R_F}{1 + |A_V|}$$

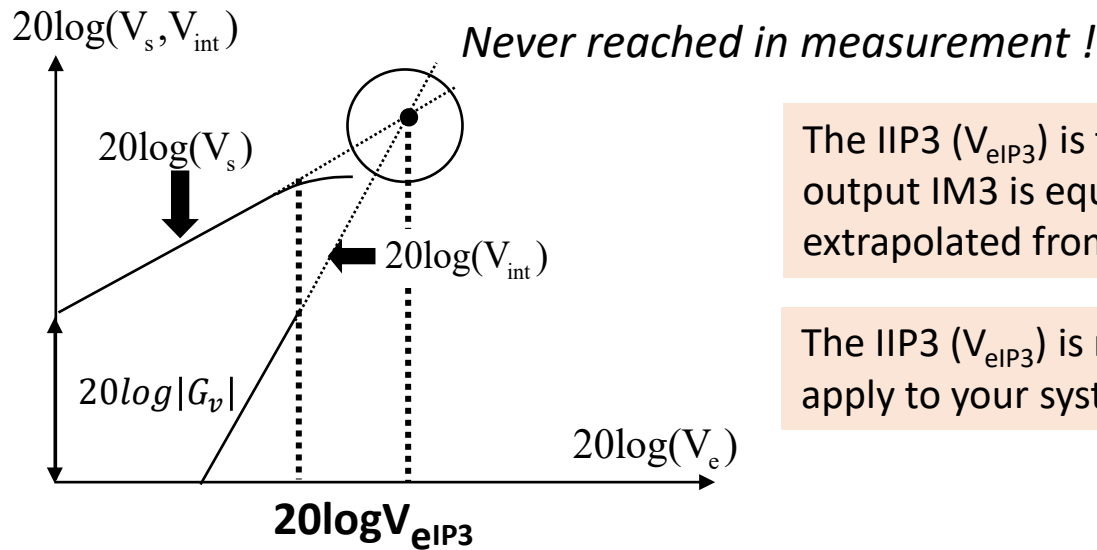
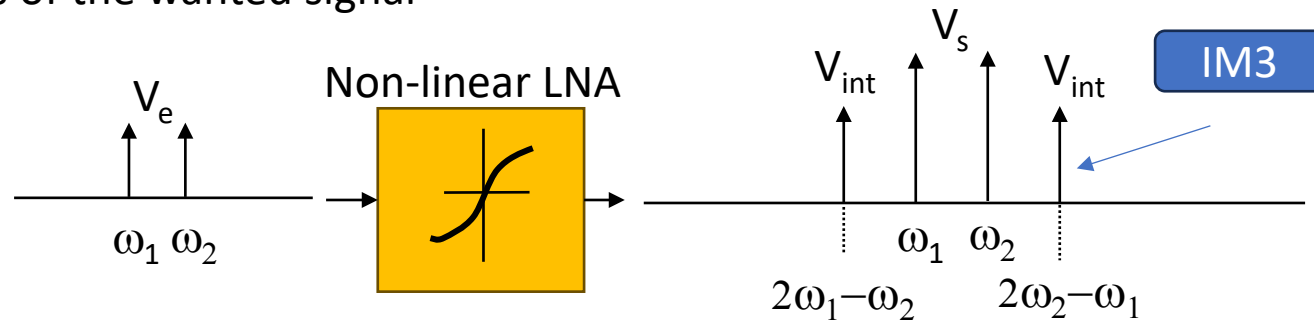


$$\Re(Z_{in}) = \frac{g_m L_1}{C_{GS}}$$

# LNA Design considerations

## Non-Linearities

$\omega_1$  and  $\omega_2$  are 2 harmonics of the wanted signal



The IIP3 ( $V_{eIP3}$ ) is the input signal level for which the output IM3 is equaling the fundamental harmonic ( $V_s$ ) extrapolated from small signal (AC).

The IIP3 ( $V_{eIP3}$ ) is related to the maximum power you can apply to your system

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### *R-Feedback general considerations*

- Simple
- Compact (No-inductors)
- Wideband
- RF allows to synthesize a real part in  $Z_{in}$
- $L_G$  and  $C_{GS}'$  will help in cancelling the imaginary part
- while controlling  $Q_{in}$

### *Input impedance*

$$Z_{IN} = L_G s + \left( \frac{1}{(C_{GS} + C'_{GS})s} // Z_P \right)$$

$$\Re(Z_P) = R_P = \frac{R_O + R_F}{1 + G_m R_O}$$

$$C_P = \frac{R_O^2 C_L (G_m R_F - 1)}{(R_O + R_F)^2}$$

$$\Re(Z_{IN}) = R_S = \frac{R_P}{(1 + Q_P^2)} = 50 \Omega$$

$$Q_P = R_P C_T \omega_0$$

$$C_T = C_{GS} + C'_{GS} + C_P$$

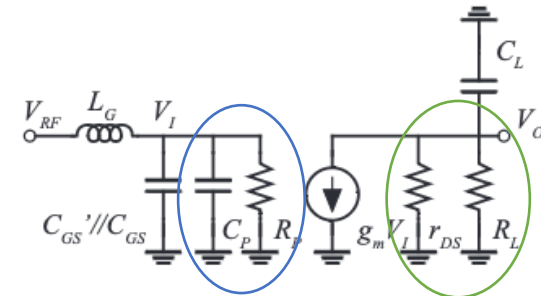
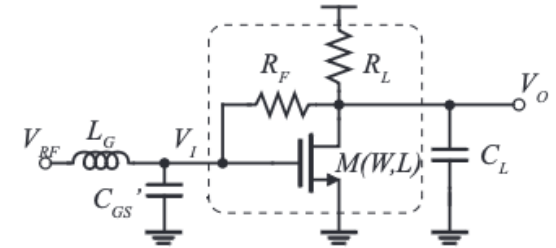
$$\Im(Z_{IN}) = 0 = L_G s + \frac{Q_P^2}{C_T s (1 + Q_P^2)}$$

**Note :**

$$\left\{ \begin{aligned} \Re(Z_P) &= \frac{1 + \left(\frac{f}{f_c}\right)^2}{\frac{1 + G_m R_O}{R_O + R_F} + \frac{1}{R_F} \left(\frac{f}{f_c}\right)^2}, \\ \Im(Z_P) &= - \left(1 + \frac{R_F}{R_O}\right)^2 \frac{1 + \left(\frac{f}{f_c}\right)^2}{2\pi f C_L (G_m R_F - 1)} \end{aligned} \right.$$

**Reduces For**

$$\frac{f_c}{f} = N > 3$$



$$R_O = \frac{R_L r_{DS}}{R_L + r_{DS}}$$

# Resistive Feedback LNA Topology

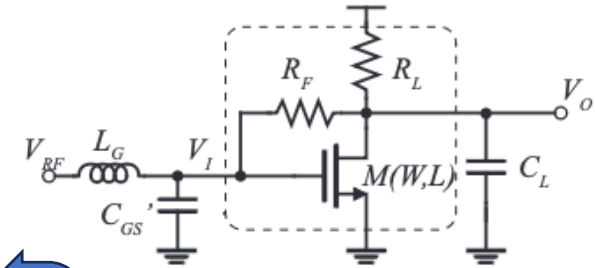
## Voltage Gain

$$|G_o| = \left| \frac{V_o}{V_{RF}} \right| = \frac{|G_T|}{\sqrt{\left(1 + \left(\frac{f_0}{f_c}\right)^2\right)}}$$

$$|G_T| = |G_v| Q_{IN} = \frac{(G_m R_F - 1) R_O}{(R_O + R_F)} \sqrt{1 + Q_P^2}$$

$$Q_{IN} = V_I / V_{RF} \quad Q_{IN} = \sqrt{1 + Q_P^2} = \sqrt{\frac{R_O + R_F}{R_S(1 + G_m R_O)}}$$

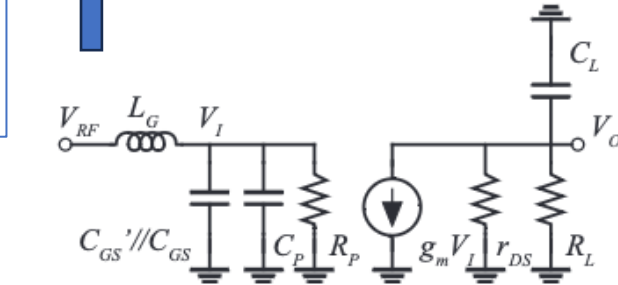
$$\frac{f_c}{f_0} = N > 3$$



## BW

$$f_c = \frac{R_O + R_F}{2\pi R_O R_F C_L} = N f_0$$

$R_S$   
(the source resistance seen  
at the 50  $\Omega$  input)



## Noise Figure

$$F = 1 + \frac{4 \left( \frac{R_F}{Q_{IN}^2} + R_S \right)^2}{Q_{IN}^2 G_m R_S \left[ \frac{R_F}{Q_{IN}^2} + R_S + \frac{G_m R_S R_F R_L}{R_F + R_L} \right]^2} \left[ \gamma + \frac{1}{G_m R_L} + \frac{\left( 1 + \frac{Q_{IN}^2 G_m R_S R_F}{R_F + Q_{IN}^2 R_S} \right)^2}{G_m R_F} \right]$$

$$\left( F_{min} = 1 + \frac{(1 + G_m) R_S^2 R_F}{R_S (1 - G_m R_F)^2} + \frac{\gamma g_m (R_S + R_F)^2}{R_S (1 - G_m R_F)^2} + \frac{(R_S + R_F)^2}{R_S R_L (1 - G_m R_F)^2} \right)$$

## IIP<sub>3</sub>

$$V_{IIP3} = \frac{2}{Q_{IN}} \sqrt{\frac{2G_m}{G_{m3}}}$$

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### *Small signal (AC) parameters*

- $G_T$ , NF, IIP3 vs ( $G_m$ ,  $G_{m3}$ ,  $G_{DS}$ )

$$g_m = \frac{\partial I_D}{\partial V_G} = \frac{I_S}{\phi_t} g_G = \frac{I_S}{n\phi_t} \frac{2(q_s - q_d) - i_d \zeta \left( \frac{q_s}{1+q_s} - \frac{q_d}{1+q_d} \right)}{1 + \zeta (q_s - q_d)}$$

$$g_{ds} = \frac{I_S}{\phi_t} g_D = \frac{I_S}{n\phi_t} \frac{2(q_s \sigma - q_d (\sigma - n)) - i_d \zeta \left( \sigma \frac{q_s}{1+q_s} - (\sigma - n) \frac{q_d}{1+q_d} \right)}{1 + \zeta (q_s - q_d)}$$

$$g_{m3} = \frac{I_S}{(n\phi_t)^3} \left\{ \frac{\frac{2q_s}{(1+q_s)^3} - \frac{2q_d}{(1+q_d)^3} - \zeta n^2 g_{G2} \left( \frac{q_s}{1+q_s} - \frac{q_d}{1+q_d} \right) - \zeta \left[ 2n g_G \left( \frac{q_s}{(1+q_s)^3} - \frac{q_d}{(1+q_d)^3} \right) + i_d \left( \frac{q_d(1-2q_d)}{(1+q_d)^5} - \frac{q_d(1-2q_d)}{(1+q_d)^5} \right) \right]}{1 + \zeta (q_s - q_d)} \right\}$$

- Valid in all region (  $q_s$  and  $q_d$  shall be explored )
- We consider only saturation

$$q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

$$g_{msat} = \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}$$

$$g_{dsat} = \frac{\sigma}{n} \frac{2I_S}{\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}$$

$$g_{msat3} = \frac{16I_S}{(n\phi_t)^3} \frac{q_s}{(q_s + 1)^3} \frac{2 - 2\zeta q_s - 3\zeta q_s^2}{(q_s + 1)^4}$$

- It reduces the exploration to only  $q_s$ .



### *Large signal (DC) parameters*

- To compute the final voltages

$$V_T = V_{T0} - \sigma(V_{SB} + V_{DB})$$

$$V_P = \frac{V_{GB} - V_T}{n}$$

$$\frac{V_P - V_{S(D)B}}{\phi_t} = q_{s(d)} - 1 + \ln q_{s(d)}$$

- Sometime we'll use  $i_f$  in the code

$$i_d = i_f - i_r \quad \longleftrightarrow \quad q_{S(D)} = \sqrt{1 + i_{f(r)}} - 1 \quad \longleftrightarrow \quad i_d = \frac{(q_s + q_d + 2)}{1 + \zeta|q_s - q_d|} (q_s - q_d)$$

$$I_D = I_S \cdot i_d$$

$$I_S = \frac{\mu C'_{ox} n (U_T)^2 W}{2 L}$$

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## General Approach :

### Requirements

- GT, NF, IIP3, IDC  Possible Tradeoff
- fo, CL, BW  No Tradeoff

### Design parameters

- ACM parameters for a fixed L
- $V_{T0}, I_S, n, \sigma, \zeta$

### Design Variables (parameters)

- $W, q_s$  are first order design variables ( $q_s$  = inversion level sets the energy efficiency and the voltages)
- RL, RF are second order design variables

### Approach :

- Explore the different tradeoff (the design space) on GT, NF, IIP3 and IDC by playing with  $W, q_s$ .
- In saturation region (only  $q_s$  is needed)

$$g_{msat} = \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \quad g_{msat3} = \frac{16I_S}{(n\phi_t)^3} \frac{q_s}{(q_s + 1)^3} \frac{2 - 2\zeta q_s - 3\zeta q_s^2}{(\zeta q_s + 2)^4} \quad g_{dsat} = \sigma \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}$$

- Circuit equations depend on  $G_m$  and ACM gives  $g_m$  (normalized)...

$$|G_T| = \frac{(G_m R_F - 1) R_O}{(R_O + R_F)} \sqrt{\frac{R_O + R_F}{R_S(1 + G_m R_O)}} \quad G_{[m;DS]x} = \mu C'_{ox} \frac{(U_T)^{2-x}}{2} \frac{W}{L} g_{[m;DS]x} \quad I_S = \mu C'_{ox} n \frac{(U_T)^2}{2} \frac{W}{L} \quad I_{SL} = \frac{I_S}{W} = \mu C'_{ox} n \frac{(U_T)^2}{2L}$$

Where we introduce  $W$  in the design space

*Reducing the variables :*

Starting from :  $|G_T| = \frac{(G_m R_F - 1) R_O}{(R_O + R_F)} \sqrt{\frac{R_O + R_F}{R_S(1 + G_m R_O)}}$   $G_T(G_m; R_0; R_F; R_S) = G_T(\underbrace{W; q_s; I_D; V_{DSAT}}_{\text{variables}}; \underbrace{V_{T0}; I_S; n; \sigma; \zeta; f_c; C_L}_{\text{parameters}})$

$$G_{[m; DS]x} = \mu C'_{ox} \frac{(U_T)^{2-x}}{2} \frac{W}{L} g_{[m; DS]x} \rightarrow g_{msat} = \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \rightarrow \boxed{G_m(q_s; W)}$$

$$\rightarrow g_{dsat} = \frac{\sigma}{n} \frac{2I_S}{\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \rightarrow \boxed{G_{DS}(q_s; W)}$$

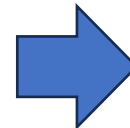
$$R_O = \frac{R_L r_{DS}}{R_L + r_{DS}} \rightarrow \boxed{G_{DS}(q_s; W)}$$

$$R_L(I_D) = \frac{V_{DD} - V_{DSAT}}{I_D} \rightarrow V_{DSAT} = \phi_t(\sqrt{1 + i_f} + 3)$$

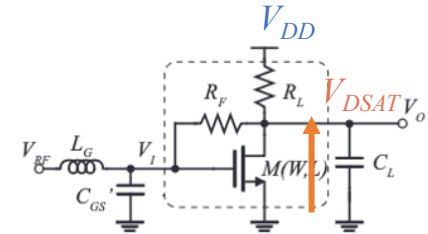
$$I_D = I_{SL} * W * i_f \quad i_f = (1 + q_s)^2 \rightarrow \boxed{R_O(q_s; W)}$$

$$R_F = \frac{R_0}{2\pi R_0 C_L f_c - 1} \rightarrow \boxed{R_F(q_s; W) \Big|_{f_c}}$$

$$\left( f_c = \frac{R_O + R_F}{2\pi R_O R_F C_L} \right)$$



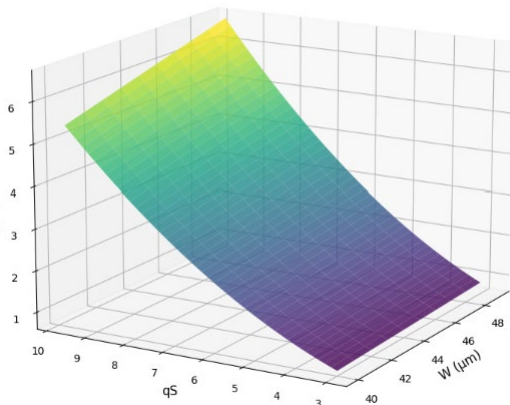
$$G_T(G_m; R_0; R_F; R_S) = G_T(W; q_s)$$



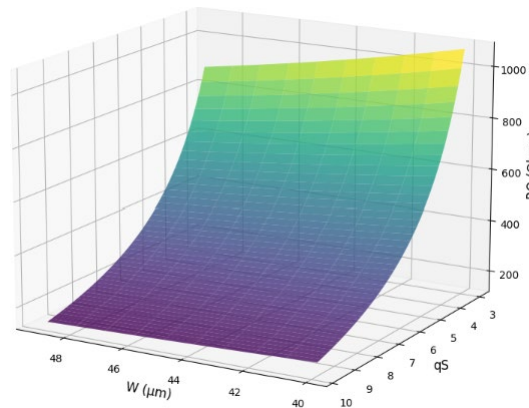
# Inversion Level Based Method for R-Feedback LNA with ACM-2

Finally :  $I_D(q_s; W)$ ;  $R_F(q_s; W)$ ;  $R_0(q_s; W)$ ;  $G_T(q_s; W)$

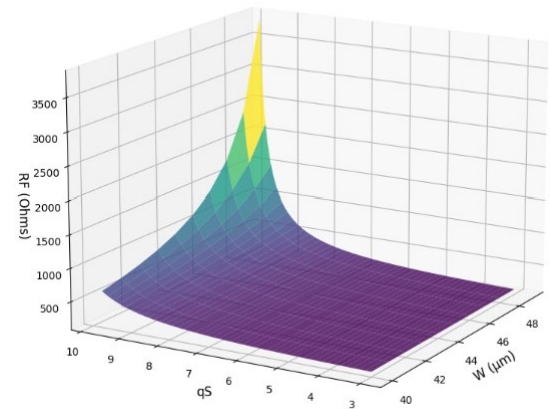
$I_D(q_s; W)$



$R_0(q_s; W)$



$R_F(q_s; W)$



*Computing  $G_T$ ,  $F$  and  $IIP3$ :*

Finally :  $|G_T| = \frac{(G_m R_F - 1) R_O}{(R_O + R_F)} \sqrt{\frac{R_O + R_F}{R_S(1 + G_m R_O)}}$

↳  $G_T(q_s; W)|_{f_c; C_L}$

$V_{IIP3} = \frac{2}{Q_{IN}} \sqrt{\frac{2G_m}{G_{m3}}}$

↳  $IIP3(q_s; W)|_{f_c; C_L; R_S}$

$$F = 1 + \frac{4 \left( \frac{R_F}{Q_{IN}^2} + R_S \right)^2}{Q_{IN}^2 G_m R_S \left[ \frac{R_F}{Q_{IN}^2} + R_S + \frac{G_m R_S R_F R_L}{R_F + R_L} \right]^2} \left[ \gamma + \frac{1}{G_m R_L} + \frac{\left( 1 + \frac{Q_{IN}^2 G_m R_S R_F}{R_F + Q_{IN}^2 R_S} \right)^2}{G_m R_F} \right]$$

↳  $F(q_s; W)|_{f_c; C_L; R_S}$

with  $G_T(q_s; W)|_{f_c; C_L}$

↳  $F(G_T; W)$

Remember :

$$Q_{IN} = V_I / V_{RF} = \sqrt{1 + Q_P^2} = \sqrt{\frac{R_O + R_F}{R_S(1 + G_m R_O)}}$$

# Inversion Level Based Method for R-Feedback LNA with ACM-2

## Exploring the design space:

Setting GT (or NF, or IIP3) helps to build 2D plots.

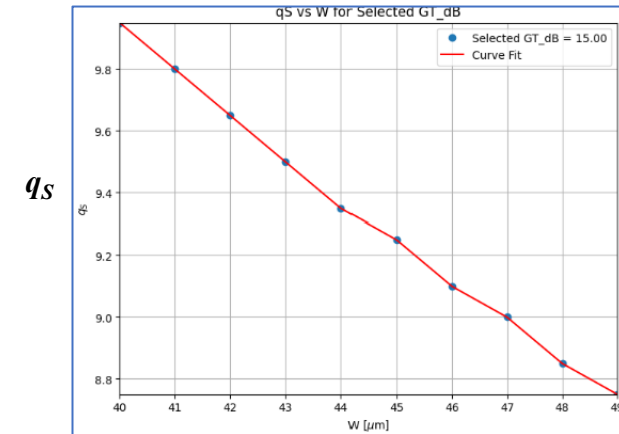
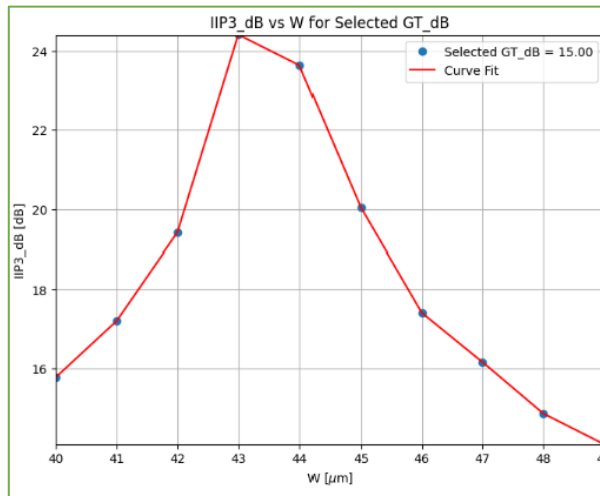
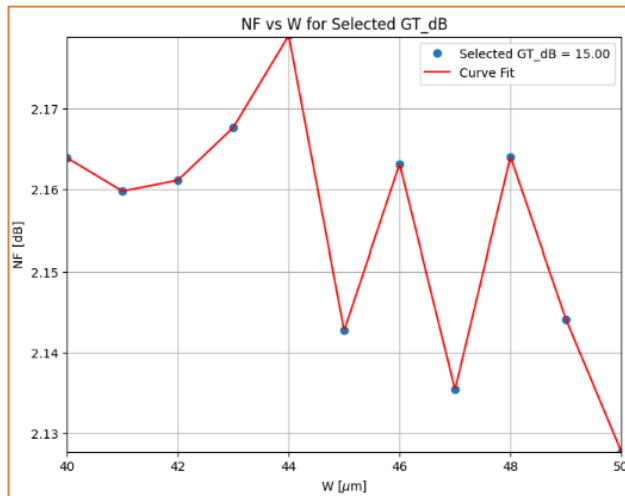
For a particular  $G_T$

=> We get the relationship  $q_s \Leftrightarrow W$

=> We plot  $NF(W)$

=> We plot  $IIP3(W)$

And We already have  $I_D$



$W(\mu m)$

## *Setting the final value:*

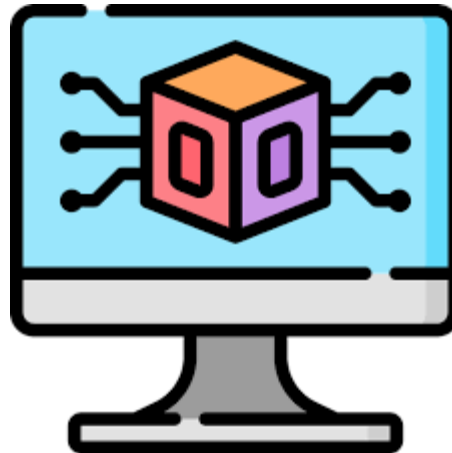
- We choose  $W$ , that gives  $q_S$  for a given  $G_T$ .
- $(W; q_S)$  gives  $R_O, R_L, I_D, G_m, V_{DSAT}$  and  $V_G$

## *Input Matching*

- With  $R_O, R_F, G_m \Rightarrow R_P$   $\Re(Z_P) = R_P = \frac{R_O + R_F}{1 + G_m R_O},$
- With  $R_S$  and  $R_P$  calculate  $Q_P$   $\Re(Z_{IN}) = R_S = \frac{R_P}{(1 + Q_P^2)} = 50 \Omega$
- With  $Q_P$  calculate  $C_T$  and  $C_{GS}'$   $Q_P = R_P C_T \omega_0 \quad C_T = C_{GS} + C_{GS}' + C_P \quad C_P = \frac{R_O^2 C_L (G_m R_F - 1)}{(R_O + R_F)^2}$
- With  $C_T$  calculate  $L_G$   $\Im(Z_{IN}) = 0 = L_G s + \frac{Q_P^2}{C_T s (1 + Q_P^2)}$



Now, let's simulate with a real PDK



# TiMA Acknowledgments



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